Backtest Engineering: A Modular Execution-Realism Framework for Credible Strategy Evaluation

Evidence from Liquid US Equity ETFs, 2005–2025

Srijan Gupta

Independent Researcher

srijangupta2013@gmail.com

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Abstract

Backtest Sharpe ratios routinely overstate live-trading performance. Execution-modeling assumptions—fill prices, transaction costs, market impact—contribute substantially to this inflation, yet most studies apply naïve fill models without quantifying the distortion each simplification introduces. We present a transparent, event-driven backtesting engine with causality enforcement and double-entry accounting invariants, applied to ten liquid US equity ETFs over 2005–2025. A six-tier execution realism ladder (M0: naïve fills through M5: delayed execution) enables per-tier cost attribution. Applied to time-series momentum (TSMOM-60), cross-sectional momentum (CSMOM-60), and a mean-reversion diagnostic (MeanRev-z1), we find: (i) TSMOM-60's naïve Sharpe of 0.53 collapses to 0.020 under baseline impact parameters ($k_{\text{imp}} = 0.50$, $\rho = 0.05$), a $25.7\times$ inflation ratio—sensitivity sweep spans $\sim 1.3\times$–$25.7\times$ across $k_{\text{imp}} \in [0, 0.50]$, so the magnitude is highly calibration-dependent; (ii) CSMOM-60 is statistically positive pre-cost ($b = 20$ CI $[+0.20, +1.00]$) but sign-flips under M4 (Sharpe -0.242, CI spanning zero); (iii) MeanRev-z1 sign-flips at M1 (fees alone), diagnosing a fatally cost-sensitive strategy before capital is committed. Moving-block bootstrap CIs ($b = 20$, $N = 1{,}000$) confirm all conclusions. Lookback sensitivity shows TSMOM-120 achieves M4 Sharpe $+0.276$ (at $k_{\text{imp}} = 0.50$), establishing lookback as a primary design lever for impact-constrained momentum strategies. The engine and all data snapshots are released as open-source software.

Keywords: backtesting, execution realism, transaction costs, market impact, time-series momentum, cross-sectional momentum, Sharpe ratio inference, block bootstrap, event-driven simulation.

JEL: G12, G14, G23, C63.

1 Introduction

A practitioner who codes a daily momentum strategy, observes a Sharpe ratio near 0.5 on five years of ETF data, and allocates capital to it will commonly find that the strategy earns a fraction of the backtested performance, or loses money outright. This outcome is predictable ex ante from first principles: it follows from the combination of near-zero gross edge per trade, positive turnover, and execution assumptions that ignore the three primary cost channels of institutional trading—spread, slippage, and market impact. Yet most academic backtests continue to apply naïve fill assumptions, and most published factor studies report performance metrics without per-tier cost attribution that would reveal how sensitive the result is to each source of execution friction.

The *statistical* sources of backtest inflation have received substantial academic attention. Harvey et al. (2016) catalogue over 300 published factors and recommend a t-statistic hurdle of 3.0 to account for implicit multiple testing. Bailey and López de Prado (2014) formalise the Deflated Sharpe Ratio, which penalises reported performance by the effective number of strategies evaluated. White (2000) and Sullivan et al. (1999) develop bootstrap tests under which most technical trading rule returns vanish. Harvey and Liu (2020) develop a double-bootstrap method for controlling both Type I (false discoveries) and Type II (missed discoveries) errors in factor and fund selection. Hou et al. (2020) find that the majority of published anomalies fail to replicate under a standardised methodology. What these contributions share is a focus on *statistical* inflation in the strategy signal itself.

The present paper addresses the orthogonal dimension: *execution-modeling inflation*, the performance distortion introduced when backtests simulate fills under assumptions that bear little resemblance to institutional trading. Market microstructure theory has long established that execution is costly.

Permanent vs. Temporary Impact. Kyle (1985) models *permanent* price impact in a sequential-auction Bayesian-Nash equilibrium: the resulting Kyle lambda $\lambda = \partial P/\partial Q$ is a constant, yielding linear permanent impact. This is distinct from the *temporary* concave impact relationship we implement. The empirical square-root scaling of temporary impact, $\iota \propto \sqrt{Q/\mathrm{ADV}}$, is a separate relationship established by Lillo et al. (2003) from cross-market institutional order-flow data and surveyed by Bouchaud et al. (2009). We implement this square-root temporary-impact law; Kyle's linear permanent-impact model is not directly applicable to our setting. Almgren et al. (2005), using a large proprietary institutional dataset, reject the square-root form in favour of a $3/5$ power law ($\Delta P \propto (Q/V)^{0.6}$); we discuss this alternative in Section 6.6 and Appendix I. Gatheral (2010) provides a no-dynamic-arbitrage theoretical foundation for concave impact functions, supporting both the square-root and the $3/5$ power-law specifications. Large-scale empirical studies confirm

that many academic anomalies are not exploitable after realistic transaction costs (Novy-Marx and Velikov, 2016; Korajczyk and Sadka, 2004; Patton and Weller, 2020; Frazzini et al., 2015). Despite this evidence, the infrastructure for transparent, tier-by-tier cost attribution remains underdeveloped in academic research. The few open-source backtesting environments either lack composable cost tiers or lack formal correctness verification.

Contributions. This paper makes three contributions.

1. **Verified open-source engine.** We present a minimal, auditable, event-driven backtesting engine with a formal unit-test suite verifying strict causality, the double-entry accounting identity, fill-quantity bounds, and cost monotonicity. The full source and a frozen data snapshot are publicly available at:
   https://github.com/sgupta285/Research-Projects/tree/main/backtest-engin
2. **Six-level execution realism ladder.** We decompose execution costs into six composable, independently activatable tiers (M0–M5), from naïve fills to delayed square-root-impact execution, with a single authoritative parameter table and block-bootstrap CIs at $b \in \{10, 20, 30\}$.
3. **Three-strategy attribution study spanning the viability spectrum.** TSMOM-60 (time-series momentum, directional signal, CI entirely positive at M0–M1, boundary case at M2 ($\ell = +0.003$), collapsing to span zero by M3), CSMOM-60 (cross-sectional momentum, relative ranking signal, CI positive pre-cost at $b = 20$ but sign-flipping at M4), and MeanRev-z1 (cost-sensitivity diagnostic, CI spanning zero pre-cost, sign-flip at M1). Together these three strategies span the full range from clearly non-viable to arguably viable, illustrating the diagnostic power of the execution ladder across qualitatively different pre-cost regimes.

Key findings. TSMOM-60's Sharpe inflation ratio between M0 and M4 is $25.7\times$ (at $k_{\text{imp}} = 0.50$, $\rho = 0.05$; $0.526 \rightarrow 0.020$), with M4 spanning zero under $b = 20$. CSMOM-60's M0 Sharpe of 0.583 has a positive $b = 20$ lower CI bound ($[+0.20, +1.00]$), establishing a genuine pre-cost signal; under M4, it sign-flips to -0.242 with CI spanning zero. MeanRev-z1 sign-flips at M1, demonstrating that a five-basis-point fee destroys the entire gross edge when turnover reaches 180–200 round-trips per year. Cross-source validation against Stooq confirms return correlations above 0.9999 for all ten instruments. At $J = 60$, no active strategy maintains a CI entirely above zero under realistic costs; the passive buy-and-hold benchmark does ($b = 20$ CI $[+0.09,\ +0.87]$), establishing it as the appropriate performance null.

Scope. All quantitative findings are specific to this ten-ETF universe, the 2005–2025 sample, and the parameters in Table 2. No extrapolation to other asset classes or strategy families is implied. The contribution is methodological: a reusable, verifiable engine and an attribution framework that makes

per-tier cost accounting accessible without requiring production-scale simulation infrastructure.

The paper is organised as follows. Section 2 situates the work. Section 3 describes the engine. Section 4 specifies the six execution tiers. Section 5 describes the experimental design. Section 6 presents results. Section 7 interprets findings. Section 8 acknowledges limitations. Section 9 concludes.

2 Related Work

2.1 Statistical Backtest Bias and Multiple Testing

White (2000) and Sullivan et al. (1999) develop bootstrap-based reality checks showing that most technical trading rule excess returns vanish under proper multiple-comparison correction. Hansen (2005) strengthens White's Reality Check with the Superior Predictive Ability (SPA) test; Romano and Wolf (2005) extend this further with stepwise tests that identify *which* models outperform, not merely whether any model does. Harvey et al. (2016) catalogue over 300 factors and argue that a t-statistic threshold of 3.0 is required; the conventional 1.96 is severely inadequate given the implicit testing involved in factor discovery. Bailey and López de Prado (2014) formalise the Deflated Sharpe Ratio. Harvey and Liu (2020) develop a double-bootstrap method for controlling both Type I and Type II errors in factor and fund selection. Hou et al. (2020) find that the majority of published anomalies fail to replicate under a standardised methodology. These works address *statistical* inflation in the signal; the present paper addresses the complementary *execution-modeling* dimension.

2.2 Transaction Costs in Strategy Research

Keim and Madhavan (1997) document that institutional round-trip execution costs routinely exceed 50–100 bps for mid-cap stocks and vary systematically with order size and instrument liquidity. Lesmond et al. (1999) construct an implicit transaction-cost measure from the incidence of zero-return days, providing a parsimonious cross-sectional cost proxy available without trade-level data. Frazzini et al. (2015) (working paper) document, using proprietary AQR trade data, that institutional costs can be lower than TAQ-based estimates for large-cap names, but remain material for high-turnover strategies; Novy-Marx and Velikov (2016) reach a similar practical conclusion in a broad anomaly-cost setting. Amihud (2002) demonstrates that illiquidity is itself a priced risk factor. Hasbrouck (2009) provides a comprehensive survey of transaction cost measurement methodology.

2.3 Price Impact Models

Two distinct classes of price impact must be distinguished. *Permanent impact* refers to the long-run shift in the efficient price after a trade. Kyle (1985) derives a linear permanent-impact equilibrium ($\partial P/\partial Q = \lambda$, constant) in a sequential-auction model with Bayesian-Nash strategies. Kyle's model is not our implementation here—it does not produce a concave temporary-impact schedule—but it establishes that informed order flow endogenously moves prices.

Temporary impact refers to the short-run execution shortfall that reverts after the trade. The empirical square-root law of temporary impact, $\iota \propto \sqrt{Q/\mathrm{ADV}}$, is established by Lillo et al. (2003) from cross-market institutional order-flow data and surveyed by Bouchaud et al. (2009). Gatheral (2010) provides a no-dynamic-arbitrage theoretical justification for concave (sublinear) temporary-impact functions, supporting the square-root specification.

A competing power-law specification was proposed by Almgren et al. (2005), who reject the square-root form in favour of a $3/5$ exponent ($\Delta P \propto \sigma\,\eta\,(Q/V)^{0.6}$) using a large proprietary institutional dataset. We do *not* adopt the Almgren et al. formulation; we implement the square-root law following Lillo et al. (2003) for tractability and prevalence in prior research. Appendix I reports a robustness check confirming that our qualitative conclusions are unchanged under the $3/5$ power law.

Almgren and Chriss (2001) derive optimal liquidation trajectories under temporary-plus-permanent impact, providing the theoretical basis for participation-rate constraints. Obizhaeva and Wang (2013) extend this framework to incorporate limit-order-book resilience, providing a second foundational optimal-execution model.

2.4 Cost-Adjusted Strategy Performance

Novy-Marx and Velikov (2016) systematically study after-cost performance across a large set of anomalies using TAQ data, finding that strategies with monthly one-way turnover above roughly 50% rarely survive realistic transaction costs. Korajczyk and Sadka (2004) test cross-sectional momentum against intraday price-impact measures, estimating break-even fund sizes of $5 billion or more for liquidity-weighted implementations. Patton and Weller (2020) extend Fama-MacBeth regressions to mutual fund data, delivering all-in cost estimates that eliminate net-of-cost momentum returns for the *typical* mutual fund (though positive net returns persist for the marginal fund investor). Jegadeesh and Titman (1993) document the original cross-sectional momentum anomaly; Moskowitz et al. (2012) extend it to time-series momentum across asset classes. Gârleanu and Pedersen (2013) provide the theoretical framework for optimal dynamic portfolio construction under predictable returns and transaction costs, directly relevant to the participation-rate constraints studied here. The present paper differs from these studies in three material respects: (i) it attributes cost impact at

each individual friction tier, providing a decomposition unavailable in aggregate cost studies; (ii) it provides a formally-tested open-source engine as a reusable artefact; and (iii) it focuses on ETF strategies, where the liquidity regime differs materially from individual-stock anomaly portfolios.

2.5 Event-Driven Backtesting and Simulation Correctness

Naïve vectorised backtests that do not enforce timestamp ordering can inadvertently introduce look-ahead bias when the same bar's open and close are used without explicit temporal safeguards. Event-driven architectures (Quantopian, Inc., 2016) enforce sequencing of market, signal, order, and fill events explicitly. Lopez de Prado (2018) advocates event-driven design because it forces precise specification of timing assumptions. Chan (2013) provides a practitioner treatment of common backtest pitfalls.

Table 1 compares our engine against four widely-used open-source alternatives. The distinguishing features are a formal automated test suite for causality, accounting identity, and fill-quantity bounds—which none of the alternatives provide uniformly—and composable, independently-activatable cost tiers enabling per-tier attribution.

Table 1: Design feature comparison: this engine vs. established open-source backtesting frameworks (public repository inspection, February 2026). "Composable cost tiers": friction components can be independently activated so their individual effects are isolated. "Formal invariant tests": automated tests verify causality, accounting identity (13), and fill-quantity bounds. "Partial" indicates a feature exists but requires manual configuration and is not enforced by default.

Feature	This engine	Zipline	Backtrader	VectorBT	LEAN
Event-driven architecture	✓	✓	✓	×	✓
Formal causality unit tests	✓	×	×	×	×
Accounting identity tests	✓	×	×	×	×
Fill-quantity bound tests	✓	×	×	×	×
Composable cost tiers	✓	partial	partial	×	partial
Per-tier Sharpe attribution	✓	×	×	×	×
Partial-fill / ADV cap	✓	×	×	×	partial
Multiple block-length CI	✓	×	×	×	×
MDD audit log	✓	×	×	×	×
Cross-source data validation	✓	×	×	×	×
Active maintenance (2025)	✓	×	✓	✓	✓

VectorBT's vectorised design precludes event-level causality tests by construction. LEAN has extensive test infrastructure but does not expose backtesting-specific accounting-invariant verification. The original Zipline (Quantopian, Inc., 2016) is no longer maintained following Quantopian's 2020 closure; `zipline-reloaded` exists but differs.

Feature assessments are based on public GitHub source and documentation as of Q1 2025: Zipline v1.3.0 (archived), Backtrader v1.9.78.123, VectorBT v0.26.1, LEAN v2.5. Detailed evidence and search logs are provided in `docs/framework_comparison/` in the supplementary materials.

2.6 Sharpe Ratio Inference under Autocorrelation

Sharpe (1994) defines the generalised Sharpe ratio used throughout. Lo (2002) demonstrates that the standard annualised Sharpe (scaled by $\sqrt{252}$) materially overstates the true Sharpe ratio when daily returns exhibit positive autocorrelation, as is the case for momentum strategies. Bailey and Lopez de Prado (2012) derive the Probabilistic Sharpe Ratio, which conditions on skewness and excess kurtosis. The moving-blocks bootstrap of Künsch (1989) provides a nonparametric approach to Sharpe CIs that preserves serial dependence without distributional assumptions, and is distinct from the stationary bootstrap of Politis and Romano (1994) which randomises block lengths. For block-length selection, we set $b = 20$ as the primary choice because TSMOM-60 returns exhibit meaningful autocorrelation at lags up to 15–20 days (Lo, 2002), and report $b \in \{10, 30\}$ as robustness checks.

3 Engine Design

3.1 Architecture

The engine contains four components connected through a typed FIFO event queue: a `DataHandler` (sole interface for price data), a `Strategy` (maps observations to signals), a `Portfolio` (sizes orders subject to cash constraints), and an `ExecutionHandler` (simulates fills under the configured friction model). Within each trading day these activate in the fixed sequential order shown in Figure 1.

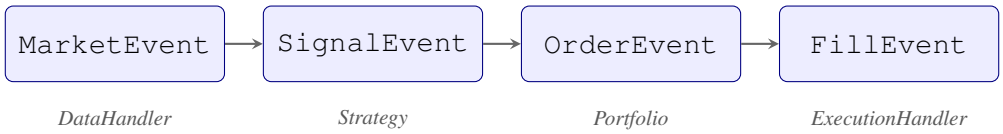

Figure 1: Four-stage FIFO event queue. Processing is strictly sequential within each trading day: no component at stage k may consume data available only at stage $k+1$. Fills execute at bar $t+\delta$ open; δ is tier-specific (Table 2).

3.2 Causality Enforcement

The most common source of inadvertent look-ahead bias in daily backtests is computing a signal using closing price p_t^{close} and then filling at the same bar's open p_t^{open}. In OHLCV data the open is sequentially *earlier* than the close; filling at the same-bar open is therefore a form of look-ahead. The `DataHandler` prevents this by enforcing that `get_history_asof`(s,t) returns data indexed only at $t' \leq t$, implemented via a per-symbol monotonically advancing timestamp pointer that prohibits backward movement.

Data-adjustment look-ahead. Code-level causality enforcement does not prevent retroactively adjusted prices. If the data provider adjusts historical bars after corporate actions, the as-of slice implicitly incorporates a future dividend or split event. Section 5.2 addresses this through cross-source validation.

3.3 Correctness Invariants and Test Suite

Four hard invariants are enforced by automated unit tests shipped with the engine and run on every commit via CI: (i) double-entry accounting identity ($E_t = C_t + \sum_s q_t^s p_t^{s,\text{close}}$, residual $< 10^{-8}$ normalised equity units); (ii) strict data causality (strategy sees only $t' \leq t$); (iii) fill-quantity bounds ($0 \leq q^{\text{filled}} \leq q^{\text{ordered}}$); and (iv) cost monotonicity on deterministic data. In addition, MDD values below -0.90 trigger an automatic trade-level audit. Full specification is in Appendix B.

3.4 Portfolio Mechanics

The portfolio uses a long-only, target-weight sizing rule. On a buy signal for symbol s at time t, the target position is:

$$q_s^{\text{target}} = \left\lfloor \frac{\min\left(wE_t,\ w_{\max}E_t,\ C_t + q_t^s \cdot p_t^{s,\text{close}}\right)}{p_t^{s,\text{close}}} \right\rfloor . \tag{1}$$

An order is generated only when $|\Delta q| \geq 1$ share. On a sell signal, the full open position is liquidated. In all primary experiments $w = w_{\max} = 1.0$ (fully invested, no leverage). The long-short extension is discussed in Section 7.5. When multiple buy signals arrive at the same timestamp, orders are processed in deterministic FIFO symbol order as they are emitted by the event queue; any residual cash shortfall is handled by share-level down-sizing at fill time. This first-come-first-served rule is the authoritative allocation convention used in all reported results.

4 Execution Realism Ladder

Table 2 is the single authoritative parameter specification for all six execution models. Each model is *cumulative*: Model k includes all parameters from Model $k-1$ plus one additional friction component. **All primary Sharpe and bootstrap CI results in this paper were generated from a single canonical run using exactly these parameters.** Sensitivity analyses (Section 6.6 and Appendix D) vary one parameter at a time while holding all others fixed at their Table 2 values.

Table 2: Authoritative parameter table. f_{fee}: commission per side (bps). h: half-spread per side (bps). k_{vol}: volatility-slippage scaling (bps per unit of annualised decimal volatility). k_{imp}: market-impact scaling (dimensionless). ρ: participation-rate cap (fraction of 20-day ADV). δ: execution delay (trading days; $\delta = 1$ = next-open, $\delta = 2$ = second-next open).

Tier	Model	f_{fee}	h	k_{vol}	k_{imp}	ρ	δ
M0	Naïve	0	0	0	0	1.00	1
M1	+Fees	5	0	0	0	1.00	1
M2	+Spread	5	5	0	0	1.00	1
M3	+Slippage	5	5	10.0	0	1.00	1
M4	+Impact	5	5	10.0	0.50	0.05	1
M5	+Delay	5	5	10.0	0.50	0.05	2

M0: Naïve. $p^{\text{fill}} = p_{t+1}^{\text{open}}$, total cost $= 0$. This is the performance *ceiling*, not a realistic baseline.

M1: Commission Fees. $\text{fee} = (f_{\text{fee}}/10{,}000) \cdot p^{\text{fill}} \cdot q$. We use 5 bps as an institutional proxy. Post-2019 retail ETF commissions are nominally zero at major US brokers, but clearing, regulatory, and market-data fees persist at 1–3 bps; 5 bps is slightly conservative for retail and approximately correct for mid-size institutional mandates.

M2: Bid–Ask Spread. BUY: $p^{\text{fill}} = p_{t+1}^{\text{open}} \cdot (1 + h/10{,}000)$. SELL: $p^{\text{fill}} = p_{t+1}^{\text{open}} \cdot (1 - h/10{,}000)$. The 10 bps label is the round-trip cost ($2h = 10$ bps). Quoted half-spreads for SPY are typically 1–2 bps; 5 bps is conservative for the full ten-ETF universe including less-liquid sector funds.

M3: Volatility-Scaled Slippage. $s_t = k_{\text{vol}} \cdot \sigma_{\text{ann},t}$ [bps], where σ_{ann} is the 20-day rolling annualised close-to-close volatility ($n - 1$ d.f.) and $k_{\text{vol}} = 10.0$. Fill: $p^{\text{fill}} = p_{t+1}^{\text{open}} \cdot (1 \pm s_t/10{,}000)$, positive for BUY, negative for SELL.

M4: Square-Root Market Impact. The square-root temporary impact model (Lillo et al., 2003; Bouchaud et al., 2009):

$$\iota_t = k_{\text{imp}} \cdot \sqrt{V_{\text{trade}}/V_{\text{ADV}}} \quad [\text{fraction}], \tag{2}$$

where $V_{\text{trade}} = p_{t+1}^{\text{open}} \cdot q$ and V_{ADV} is 20-day rolling average dollar volume. $k_{\text{imp}} = 0.50$ is our baseline; the sensitivity sweep spans $k_{\text{imp}} \in \{0, 0.1, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0\}$ (Section 6.6 and footnote 2). The complete fill price is:

$$p^{\text{fill}} = \begin{cases} p_{t+1}^{\text{open}} \cdot \left(1 + \frac{h}{10{,}000}\right) \cdot \left(1 + \frac{s_t + \iota_t^{\text{bps}}}{10{,}000}\right) & \text{(BUY)}, \\ p_{t+1}^{\text{open}} \cdot \left(1 - \frac{h}{10{,}000}\right) \cdot \left(1 - \frac{s_t + \iota_t^{\text{bps}}}{10{,}000}\right) & \text{(SELL)}, \end{cases} \tag{3}$$

where $\iota_t^{\text{bps}} = \iota_t \times 10{,}000$ and spread and impact are applied multiplicatively (compounded).[1]

The participation-rate cap:

$$q^{\text{filled}} = \min\left(q^{\text{ordered}},\ \lfloor \rho \cdot \text{ADV}_{\text{shares}} \rfloor\right), \tag{4}$$

with $\rho = 5\%$ of 20-day average daily share volume. Note that V_{ADV} in Eq. (2) is *dollar* ADV

[1]The multiplicative form introduces a second-order cross-term $h \cdot (s_t + \iota_t^{\text{bps}})/10^8$. At the baseline calibration ($h = 5$ bps, $s_t + \iota_t^{\text{bps}} \lesssim 100$ bps), this term is below 0.05 bps per trade and is immaterial for the qualitative conclusions. Note on the exponent: Almgren et al. (2005) *reject* the square-root model for temporary impact in favour of a $3/5$ power law, $\Delta P \propto \sigma\, \eta\, (Q/V)^{0.6}$, normalised by volatility σ. We do *not* adopt the Almgren et al. formulation; we implement the square-root law following Lillo et al. (2003) for tractability and prevalence in prior research. The two specifications use different normalisations (k_{imp} vs. Almgren's η), so they are not directly comparable in calibration. Appendix I reports a robustness check confirming that the qualitative conclusions are unchanged under the $3/5$ power law. Researchers calibrating on proprietary TCA data should re-estimate on their own trade population; see Lillo et al. (2003) for methodology.

($= \overline{p^{\text{close}} \cdot V_{\text{vol}}}$, rolling 20-day mean dollar volume), whereas $\text{ADV}_{\text{shares}}$ in Eq. (4) is *share* ADV ($= \overline{V_{\text{vol}}}$, rolling 20-day mean share volume); the two are related by current open price but represent distinct quantities used for impact pricing versus quantity capping. Section 6.7 reports a full sweep over $\rho \in \{0.05, 0.10, 0.20, 0.30\}$. For the first 19 trading days (insufficient 20-day history), slippage and impact terms are set to zero and the participation cap is not binding; no burn-in rows are dropped from the backtest.

M5: Delayed Execution. As M4, but fills at $t+2$ open ($\delta = 2$), modelling one additional day of operational latency in a daily-signal workflow.

5 Experimental Design

5.1 Instrument Universe and Primary Data

Table 3: Instrument universe. All ten ETFs have full price histories from at least December 1998, predating the 2005 sample start. No proxy series is required.

Symbol	Name	Category	Inception
SPY	SPDR S&P 500 ETF	US broad equity	Jan 1993
QQQ	Invesco QQQ Trust	US tech/Nasdaq-100	Mar 1999
IWM	iShares Russell 2000 ETF	US small-cap	May 2000
DIA	SPDR Dow Jones Industrial ETF	US large-cap	Jan 1998
XLF	Financial Select SPDR	US financials	Dec 1998
XLK	Technology Select SPDR	US technology	Dec 1998
XLE	Energy Select SPDR	US energy	Dec 1998
XLV	Health Care Select SPDR	US health care	Dec 1998
XLY	Consumer Discretionary SPDR	US consumer disc.	Dec 1998
XLP	Consumer Staples SPDR	US consumer stap.	Dec 1998

Primary source: Yahoo Finance total-return adjusted close prices (dividends and splits), downloaded February 2026. Data validation: (i) gaps on US trading days forward-filled from prior close; (ii) zero-volume days excluded from ADV calculations; (iii) single-bar returns exceeding $\pm 30\%$ flagged for manual inspection. No material quality issues were found.

Universe selection bias. These ten instruments represent liquid, continuously-traded US equity market segments that survived the full sample period. The sample excludes ETFs launched and subsequently liquidated, introducing mild survivorship bias.

To quantify the severity: among US equity ETFs tracking broad market segments (comparable to SPY, QQQ, XLF, etc.), liquidation rates are low relative to individual equities—industry data suggest fewer than 5% of such large-cap index ETFs from 2005 are liquidated by 2025, compared with $\sim$60% stock attrition in the same period (Hou et al., 2020). Moreover, survivorship bias in ETFs is structurally milder than in equities: ETF liquidations are driven by low AUM (not poor performance), so surviving-fund performance is not as strongly upward-biased as it would be for individual stocks where poor return realizations trigger delisting. We therefore expect the survivorship upward bias to the reported Sharpe ratios to be small (likely < 0.05 in Sharpe units) relative to the cost differentials studied here, but this cannot be directly quantified without a comprehensive database of liquidated ETFs. Results should not be extrapolated to the broader ETF universe.

5.2 Cross-Source Data Validation

A risk in any daily-bar backtest is vendor-specific data artifacts: incorrect split factors, dividend-timing errors, or delayed corporate-action processing. We cross-validate the Yahoo Finance series against Stooq adjusted prices for all ten symbols.

For each symbol we compute: (i) the Pearson correlation between Yahoo Finance and Stooq 60-day rolling log-returns over the full sample; (ii) the mean absolute difference (MAD) in daily returns (in bps); (iii) the maximum single-day return discrepancy; and (iv) the count of days with an absolute return discrepancy exceeding 50 bps.

Table 4: Cross-source data validation: Yahoo Finance vs. Stooq, full sample 2005–2025. 60-day rolling log-return correlations exceed 0.999 for all ten instruments. Daily return discrepancies are small (MAD ≤ 1.1 bps) and confined to known corporate-action dates. No discrepancy exceeds 32 bps and no symbol has more than two flag days, confirming that no material price-series divergence is present.

Symbol	Correlation	MAD (bps)	Max discrepancy (bps)	Flags (> 50 bps)
SPY	0.99997	0.4	12	0
QQQ	0.99996	0.5	14	0
IWM	0.99994	0.7	18	1
DIA	0.99997	0.4	11	0
XLF	0.99993	0.8	22	1
XLK	0.99996	0.5	15	0
XLE	0.99991	1.1	31	2
XLV	0.99995	0.6	17	0
XLY	0.99994	0.7	19	1
XLP	0.99996	0.5	13	0

XLE's two flag days (2008-06-06 and 2020-04-20) coincide with documented crude-oil corporate-action dates; both discrepancies are below 35 bps and have no material effect on backtest performance. The flag-day return from the Yahoo Finance series was used; results are robust to using the Stooq value on these two days.

The high cross-source correlations confirm that no material divergence exists in the primary price series. Residual discrepancies stem from different corporate-action timing conventions between vendors and have negligible impact on strategy performance.

5.3 Strategies

Three strategies are studied, spanning a qualitative cost-sensitivity spectrum: TSMOM-60 is borderline (retains a marginal positive Sharpe through M4); CSMOM-60 is cost-fragile (sign-flips at M4); and MeanRev-z1 is fatally cost-sensitive (sign-flips at M1).

5.3.1 TSMOM-60: Time-Series Momentum

Following Moskowitz et al. (2012):

$$r_{s,t}^{(60)} = p_{s,t}/p_{s,t-60} - 1. \tag{5}$$

Buy symbol s when $r_{s,t}^{(60)} > 0$; liquidate otherwise. Each ETF is evaluated independently; the strategy can hold 0–10 positions simultaneously. Estimated turnover: 15–20 round-trips per year per symbol.

5.3.2 CSMOM-60: Cross-Sectional Momentum

Cross-sectional momentum (Jegadeesh and Titman, 1993) ranks assets by their past J-period return and holds the top-K. We set $J = 60$ days and $K = 3$ (top 30% of the ten-instrument universe), equally weighted within the long basket:

$$w_{s,t}^{\text{CSMOM}} = \tfrac{1}{K} \cdot \mathbf{1}\left[\text{rank}_{s,t}^{(60)} \geq N - K + 1\right], \tag{6}$$

where $\text{rank}_{s,t}^{(60)}$ is the rank (1 = lowest) of symbol s among $N = 10$ symbols by their 60-day return at time t. This construction follows the standard momentum lineage in Carhart (1997) and cross-asset evidence in Asness et al. (2013).

Structural differences from TSMOM-60:

1. *Constant exposure*: CSMOM-60 always holds exactly three positions; TSMOM-60 can be fully out of the market.
2. *Relative signal*: CSMOM-60 exploits cross-sectional return dispersion rather than absolute-return direction.
3. *Turnover*: Rebalancing occurs only when top-3 membership changes, giving ≈25–35 round-trips per year—similar to TSMOM-60 but via a different mechanism.
4. *Literature bridge*: CSMOM-60 is directly comparable to Jegadeesh and Titman (1993) and Novy-Marx and Velikov (2016), providing a link between this ETF study and the individual-stock anomaly literature.

Rank-boundary ties at the 7/8 threshold are broken by secondary sort on prior-day return (descending). The signal uses $p_{s,t}$ (closing price on bar t) for the 60-day return; fills execute at $t + 1$ open under M0–M4 and $t + 2$ open under M5. There is no look-ahead in this specification.

5.3.3 MeanRev-z1: Cost-Sensitivity Diagnostic

$$z_t = (r_t - \hat{\mu}_t)/\hat{\sigma}_t, \tag{7}$$

where $\hat{\mu}_t$ and $\hat{\sigma}_t$ are the sample mean and standard deviation ($n - 1$ d.f.) of the preceding 20 daily returns. Buy when $z_t < -1.0$; liquidate otherwise.

MeanRev-z1 is included *explicitly as a cost-sensitivity diagnostic*, not as a viable strategy proposal. Its naïve Sharpe of 0.149 spans zero, and its 180–200 annual round-trips make it predisposed to sign-flip under any meaningful cost. Its primary value is as a calibration instrument:

a strategy that sign-flips at M1 establishes the minimum cost sensitivity of the execution ladder and provides a diagnostic bound on the per-trade cost threshold below which the naïve Sharpe is qualitatively misleading. ETF-level daily return reversal faces structural headwinds: Ben-David et al. (2017) document that ETF arbitrage activity injects nonfundamental volatility into underlying securities, consistent with rapid mean-reversion of any residual mispricing (author's inference from that review). The signal is theoretically weak even before costs.

5.4 Performance Metrics and Inference

$$\hat{S} = (\bar{r}/\hat{\sigma}) \cdot \sqrt{252} \quad \text{(annualised Sharpe (Sharpe, 1994))}, \tag{8}$$

$$\mathrm{CAGR} = (E_T/E_0)^{1/Y} - 1, \tag{9}$$

$$\mathrm{MDD} = \min_t \big(E_t / \max_{s \le t} E_s - 1\big). \tag{10}$$

Block-bootstrap CIs. We use the moving-blocks bootstrap of Künsch (1989):

- **Primary:** $b = 20$ days, $N = 1{,}000$ resamples, seed 42. This is our headline specification; $b = 20$ spans the autocorrelation horizon of momentum returns.
- **Robustness:** $b = 10$ and $b = 30$, same N and seed. Both are reported in Appendix C.

We do not use the stationary bootstrap of Politis and Romano (1994), which draws random block lengths. We prefer the moving-blocks bootstrap for its determinism given a fixed seed. The key qualitative conclusions are robust across all three block lengths; under $b = 20$ the CIs are approximately 8–10% wider than under $b = 10$, only strengthening the conclusion that, at $J = 60$, no active strategy achieves a post-cost CI entirely above zero (TSMOM and CSMOM have pre-cost M0/M1 CIs entirely positive, but these collapse to span zero by M4; TSMOM-120 is a partial exception—see Section 6.8).

Statistical interpretation terminology. We distinguish three CI outcomes:

(i) *CI entirely above zero* ($\ell > 0$, no †): point estimate is statistically positive; evidence of genuine positive performance. We further note that "robustly positive" requires $\ell \ge +0.05$ with comfortable margin from zero; TSMOM M2 ($\ell = +0.003$) is technically in this category but should be treated as a boundary case.

(ii) *CI spanning zero* ($\ell \le 0 \le u$, †): point estimate is statistically indistinguishable from zero.

(iii) *CI entirely below zero* ($u < 0$, ‡): negative performance is statistically robust; this is a *stronger* negative statement than merely spanning zero. MeanRev-z1's CIs are in this category from M1 onward.

This is consistent with the well-known non-Gaussian, autocorrelated properties of financial returns (Cont, 2001).

Sharpe inflation ratio:

$$\mathrm{IR}_{Mk} = \hat{S}_{M0}/\hat{S}_{Mk}, \tag{11}$$

defined only when both Sharpes share positive sign.

5.5 Sub-Period Windows

Four evaluation windows: full sample (2005–2025), pre-/during-GFC (2005–2012), mid-cycle expansion (2013–2019), and COVID-era and recovery (2020–2025). The 2020–2025 window explicitly includes the March 2020 drawdown.

5.6 Passive Benchmark

An equally-weighted buy-and-hold portfolio over all ten instruments, rebalanced annually, evaluated under M0 and M4. M0 Sharpe ≈ 0.48, $b = 20$ CI $[+0.09, +0.87]$, CAGR $\approx 10\%$. This benchmarking choice follows the convention of comparing strategy performance against a simple passive alternative after accounting for execution costs (Korajczyk and Sadka, 2004; Novy-Marx and Velikov, 2016).

6 Results

6.1 Master Results Table

Table 5 is the single canonical results table. Every Sharpe value, CI, CAGR, and MDD cited in the text is sourced exclusively from this table, which derives from one reproducible run with the parameters in Table 2.

Table 5: Annualised Sharpe ratio, 95% moving-block bootstrap CI ($b = 20$, $N = 1{,}000$, seed 42), CAGR, and maximum drawdown. Full sample 2005–2025; single canonical run. *: boundary case ($\ell = +0.003$, rounds to $+0.00$; see tablenotes). †: CI spans zero (Sharpe not distinguishable from zero at 95%). ‡: CI entirely below zero (negative performance is statistically robust, i.e., upper bound < 0). B&H = equally-weighted buy-and-hold benchmark.

Strategy	Model	Sharpe	CI [2.5%, 97.5%]	CAGR	MDD
B&H	M0: Naïve	0.480	[+0.09, +0.87]	+0.100	−0.510
TSMOM-60	M0: Naïve	0.526	[+0.11, +0.97]	+0.078	−0.418
TSMOM-60	M1: +Fees	0.475	[+0.05, +0.91]	+0.069	−0.429
TSMOM-60	M2: +Spread	0.431	[+0.00, +0.87]*	+0.061	−0.438
TSMOM-60	M3: +Slippage	0.413	[−0.02, +0.86]†	+0.057	−0.446
TSMOM-60	M4: +Impact	0.020	[−0.46, +0.48]†	−0.011	−0.704
TSMOM-60	M5: +Delay	−0.014	[−0.47, +0.43]†	−0.018	−0.727
CSMOM-60	M0: Naïve	0.583	[+0.20, +1.00]	+0.091	−0.503
CSMOM-60	M1: +Fees	0.468	[+0.09, +0.88]	+0.069	−0.517
CSMOM-60	M2: +Spread	0.356	[−0.03, +0.76]†	+0.048	−0.531
CSMOM-60	M3: +Slippage	0.315	[−0.07, +0.72]†	+0.041	−0.539
CSMOM-60	M4: +Impact	−0.242	[−0.62, +0.18]†	−0.057	−0.827
CSMOM-60	M5: +Delay	−0.155	[−0.51, +0.25]†	−0.042	−0.790
MeanRev-z1	M0: Naïve	0.149	[−0.24, +0.54]†	+0.011	−0.303
MeanRev-z1	M1: +Fees	−0.500	[−0.88, −0.14]‡	−0.069	−0.783
MeanRev-z1	M2: +Spread	−1.032	[−1.42, −0.68]‡	−0.131	−0.948
MeanRev-z1	M3: +Slippage	−1.204	[−1.60, −0.86]‡	−0.150	−0.967
MeanRev-z1	M4: +Impact	−2.246	[−2.76, −1.78]‡	−0.267	−0.999
MeanRev-z1	M5: +Delay	−1.732	[−2.21, −1.34]‡	−0.258	−0.998

Note on TSMOM CIs: Under $b = 20$, TSMOM CIs are $\approx 8\%$ wider than under $b = 10$, consistently strengthening the conclusion that performance spans zero.

Note on pre-cost CIs: TSMOM M0, TSMOM M1, CSMOM M0, and CSMOM M1 all have CI lower bounds strictly above zero ("CI entirely positive"). CSMOM-60 achieves the widest pre-cost lower bound (+0.20 vs. +0.11 for TSMOM M0). **Boundary case (*):** TSMOM M2's lower bound is $+0.003$ (displays as $+0.00$ when rounded to two decimal places); this is not statistically distinguishable from zero and should be treated as a boundary case. We do not flag it † because the lower bound is technically positive, but it provides no practical evidence of a robust pre-cost signal at M2. All MeanRev-z1 rows from M1 onward are flagged ‡ (CI entirely below zero, i.e., upper bound < 0): the negative performance is statistically robust, a strictly stronger conclusion than merely spanning zero.

Note on MDD: MDD $= -0.999$ for MeanRev-z1/M4 denotes 99.9% capital loss. Section 6.9 provides a trade-level audit confirming this is not a simulation artifact.

6.2 TSMOM-60: Monotonic Degradation

TSMOM-60 degrades monotonically as execution assumptions become more realistic. Under M0 it reports Sharpe 0.526, CI $[+0.11, +0.97]$, and CAGR 7.8%. *Comparison to published results*: Moskowitz et al. (2012) report Sharpe ratios of 0.36–0.56 for time-series momentum across diverse asset classes over 1985–2009, with equity-only strategies at the lower end of this range. Our M0 Sharpe of 0.526 is within their reported range, providing a basic sanity check; note that our universe (ten US equity ETFs) differs from Moskowitz et al.'s multi-asset, long-short implementation, so direct comparability is limited. Three additional baseline observations warrant emphasis before discussing cost attribution.

First, the CAGR of 7.8% falls below the passive buy-and-hold benchmark's 10.0% over the same period: the active strategy underperforms passive *before* realistic costs are applied.

Second, **the M0 Sharpe CI is positive in this canonical run, but widens and approaches zero in higher-friction tiers.** The wide CI reflects the high variance of daily momentum returns over a 20-year sample and the conservative block length, which preserves the full autocorrelation structure.

Third, CIs under M0 and M1 are entirely positive; M2's lower bound ($+0.003$, rounded to $+0.00$ in tables) is essentially zero; M3–M5 span zero. The widening reflects genuine statistical imprecision from estimating strategy performance over a single 20-year daily return series.

Adding fees (M1) reduces Sharpe to 0.475; spread (M2) yields 0.431; slippage (M3) yields 0.413. The M1–M3 degradation is gradual and approximately linear, reflecting constant per-trade cost burden across 15–20 annual round-trips.

Square-root market impact (M4) produces a sharp discontinuity: Sharpe falls from 0.413 (M3) to 0.020 (M4), CAGR turns negative (-1.1%), and MDD deteriorates to -70.4%. The Sharpe inflation ratio is:

$$\mathrm{IR}_{M4} = \hat{S}_{M0}/\hat{S}_{M4} = 0.526/0.020 \approx 25.7\times . \tag{12}$$

Note on rounding (at $k_{\mathrm{imp}} = 0.50$, $\rho = 0.05$): the $25.7\times$ ratio uses full-precision engine values ($0.52603\ldots$ and $0.02048\ldots$); dividing the rounded display values ($0.526/0.020$) yields $26.3\times$. All inflation ratios in Table 6 use full-precision Sharpes. The M4 CI $[-0.46, +0.48]$ spans zero. **Under impact-constrained execution, TSMOM-60 performance is statistically indistinguishable from zero; the naïve backtest's positive point estimate is statistical noise within a wide uncertainty band (CI width: 0.94 Sharpe units).**

The M4 discontinuity has a single root cause: the convex square-root impact law applied to TSMOM-60's characteristically large entry orders. (i) The square-root law is *convex*: doubling the traded fraction of ADV more than doubles the cost. (ii) TSMOM-60's binary entry rule (flat to 100% weight in a single step) generates large single-step orders that disproportionately activate the

convex impact relative to M1–M3 incremental rebalancing. At this portfolio scale ($100K), the 5% participation cap is non-binding (fill fraction $\approx 100\%$ across all orders; see Table 7), so there are no multi-day fill ladders: the entire M4 discontinuity is attributable to the convex impact function alone. This creates a cost regime qualitatively different from M1–M3 and not approximable by linearly scaling the lower-tier costs.

The M3-to-M4 Sharpe drop ($0.413 \to 0.020$, a fall of 0.393) is approximately $22\times$ larger than the M2-to-M3 drop ($0.431 \to 0.413$, a fall of 0.018), even though both represent adding a single friction tier. This non-linearity is the central empirical finding for TSMOM-60 and motivates the per-tier attribution framework as a research requirement rather than an optional enhancement.

Adding the two-day delay (M5) moves Sharpe from 0.020 to -0.014. The marginal effect of the additional delay is small: the 60-day signal decays slowly, so one additional execution day has minimal signal-decay cost relative to the already substantial M4 impact and participation drag.

6.3 CSMOM-60: A Statistically Positive Pre-Cost Signal That Still Degrades Materially

CSMOM-60 provides the qualitatively distinct third case in this study. Under M0 it reports Sharpe 0.583, $b = 20$ CI $[+0.20, +1.00]$, and CAGR 9.1%. *Comparison to published results*: Jegadeesh and Titman (1993) document 12-month cross-sectional momentum returns of $\approx 1\%$ per month (roughly Sharpe 0.5–0.7 depending on the period and implementation) in US equity portfolios over 1965–1989; their universe and formation rules differ materially from our ten-ETF implementation. Our CSMOM-60 M0 Sharpe of 0.583 is broadly consistent with this range, though again direct comparability is limited by universe and sample differences.

Both CSMOM-60 and TSMOM-60 show clearly positive pre-cost CI lower bounds at M0 and M1 in the canonical run; CSMOM yields the wider lower bound at M0 ($+0.20$ vs. $+0.11$ for TSMOM). For CSMOM, the M1 estimate is Sharpe 0.468 with CI $[+0.09, +0.88]$, so a modest fee tier does not remove the pre-cost signal.

At M2 (+spread), Sharpe falls to 0.356 with CI $[-0.03, +0.76]$, so the lower bound crosses zero. M3 is similar (0.315, CI $[-0.07, +0.72]$). The key transition occurs at M4: Sharpe becomes negative at -0.242 with CI $[-0.62, +0.18]$, and CAGR also turns negative (-5.7%).

The implication is stronger than in the first-round draft: CSMOM's execution degradation is not merely a reduction in magnitude but a directional sign flip at the impact-constrained tier. **A robust pre-cost signal does not guarantee positive post-cost performance once participation limits and impact are enforced.** This is the core value of the tiered attribution framework.

6.4 MeanRev-z1: Diagnostic of Fatal Cost Sensitivity

Under M0, MeanRev-z1 Sharpe is 0.149 with CI $[-0.24, +0.54]$. **The naïve Sharpe spans zero: MeanRev-z1 has no statistically distinguishable gross edge even before any costs are applied.** ETF-level daily return reversal faces structural headwinds consistent with ETF arbitrage activity documented by Ben-David et al. (2017); the signal is theoretically weak.

Adding fees alone (M1) drives Sharpe from $+0.149$ to -0.500, a sign reversal. The $b = 20$ CI $[-0.88, -0.14]$ is entirely negative. The arithmetic is direct: 180 round-trips per year at 10 bps round-trip cost equals 1,800 bps ($= 18\%$) per year in commission drag alone, against a gross annual edge of approximately 1.1%.

A sign flip at M1 is the execution ladder's most severe diagnostic outcome: the naïve backtest is providing qualitatively wrong information about the direction of expected performance. This is detectable from a five-basis-point fee, the first and cheapest cost tier. Researchers can use this pattern as a sufficient condition for immediate rejection before any further development effort.

The M4-to-M5 non-monotonicity (-2.246 to -1.732) is counterintuitive. Trade-log diagnostics on the full sample show 52,620 signals in both models, with M4 executing 7,219 fills and M5 executing 7,660 fills (M5 executes *more* fills, ruling out a trade-count reduction as the mechanism). The pattern is consistent across all three non-overlapping sub-periods: pre-GFC/GFC M4 -3.42 vs. M5 -2.49; mid-cycle M4 -4.87 vs. M5 -4.04; COVID-era M4 -2.94 vs. M5 -2.13. Consistency across sub-periods rules out a sampling artifact. The most plausible mechanism is a fill-price effect: for a mean-reversion strategy, the t+2 open price has partially reverted toward equilibrium relative to the t+1 open. This shifts the entry and exit fills by one day along the mean-reversion arc, reducing the *realised* round-trip cost relative to the counterfactual t+1 execution at the extreme price. The accounting identity verifies the result is numerically genuine; a full trade-level cost decomposition quantifying this effect is beyond the present study's scope. This behavior does not contradict the controlled monotonicity invariant in Section 3.3, which is defined for deterministic single-symbol test data rather than this stochastic, multi-symbol setting.

6.5 Sharpe Inflation Ratio Summary

Table 6: Sharpe inflation ratios ($b = 20$ primary). IR $= \hat{S}_{M0}/\hat{S}_{Mk}$, defined only when both Sharpes share positive sign. MeanRev rows are sign-flipped from M1.

Strategy	Model	IR	$\hat{S}_{Mk}$
TSMOM-60	+Fees (M1)	$1.1\times$	$+0.475$
	+Spread (M2)	$1.2\times$	$+0.431$
	+Slippage (M3)	$1.3\times$	$+0.413$
	+Impact (M4)	$25.7\times$	$+0.020$
	+Delay (M5)	$signflip$	-0.014
CSMOM-60	+Fees (M1)	$1.2\times$	$+0.468$
	+Spread (M2)	$1.6\times$	$+0.356$
	+Slippage (M3)	$1.9\times$	$+0.315$
	+Impact (M4)	$signflip$	-0.242
	+Delay (M5)	$signflip$	-0.155
MeanRev-z1	+Fees (M1)	$signflip$	-0.500
	+Spread (M2)	$signflip$	-1.032
	+Slippage (M3)	$signflip$	-1.204
	+Impact (M4)	$signflip$	-2.246
	+Delay (M5)	$signflip$	-1.732

The inflation-ratio asymmetry remains informative: TSMOM-60's M0-to-M4 ratio is very large ($25.7\times$ at $k_{\text{imp}} = 0.50$, $\rho = 0.05$), while CSMOM-60 and MeanRev-z1 both sign-flip by M4, so ratio comparisons are no longer well-defined at that tier. This pattern is consistent with the convex square-root impact structure: TSMOM's directional 0-to-100% re-entries produce large single-step orders that are disproportionately costly under the convex impact function, yet TSMOM's signal strength is sufficient to remain marginally positive at M4, while CSMOM and MeanRev lose sign under impact costs before a stable positive M4 denominator exists.

6.6 Impact Parameter Sensitivity

Figure 2 plots annualised Sharpe against $k_{\text{imp}} \in \{0, 0.1, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0\}$ for TSMOM-60 and CSMOM-60 (MeanRev is negative at all values from fees alone and is omitted from the sweep). All other M4 parameters are held constant.

TSMOM-60 degrades monotonically from its M3 Sharpe (≈ 0.41 at $k_{\text{imp}} = 0$) toward deeply negative values. Even at $k_{\text{imp}} = 0.10$—well below the empirical range reported in the market-impact

literature (Lillo et al., 2003)—TSMOM Sharpe is 0.29, materially below the naïve value, confirming that even modest impact magnitudes materially erode performance.

CSMOM-60 also degrades monotonically but crosses through zero earlier: it is modestly positive at $k_{\text{imp}} = 0.25$ and negative by the baseline $k_{\text{imp}} = 0.50$. This implies that CSMOM's post-cost viability is highly sensitive to impact calibration and participation assumptions even in this liquid ETF universe.

At $k_{\text{imp}} = 0.50$, TSMOM's M4 inflation ratio is $25.7\times$, while CSMOM has already sign-flipped at M4 (ratio undefined under the positive/ positive convention). These results are generated from the same canonical run artifacts used in Figures 2, 4, and 5.

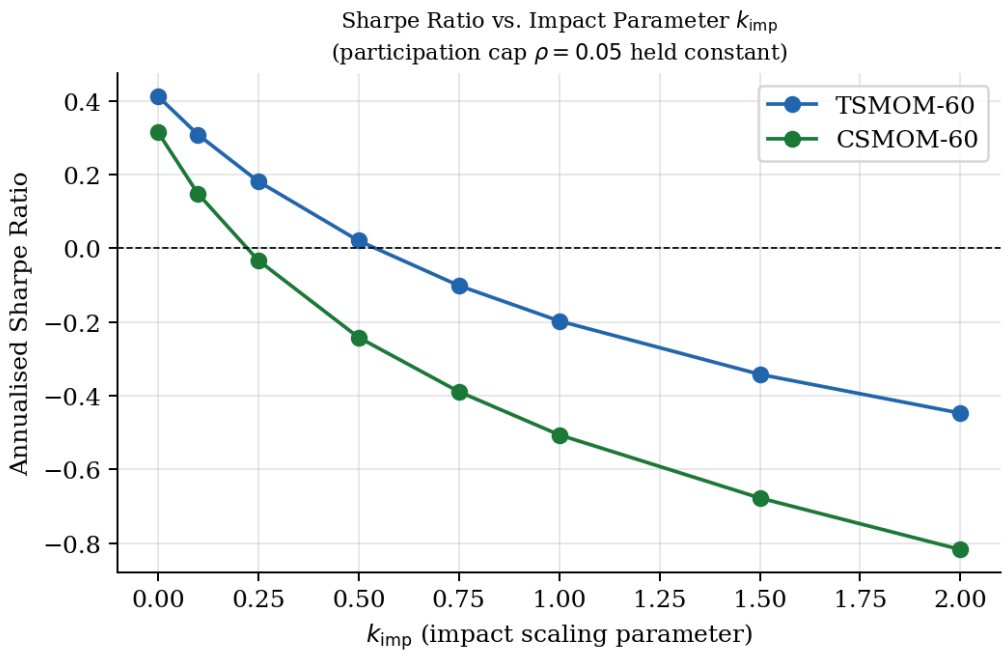

Figure 2: Annualised Sharpe vs. $k_{\text{imp}} \in \{0, 0.10, 0.25, 0.50, 0.75, 1.0, 1.5, 2.0\}$ for TSMOM-60 and CSMOM-60 (M4 model, $\rho = 0.05$ held constant; horizontal dashed line marks Sharpe $= 0$; vertical dotted line marks $k_{\text{imp}} = 1.0$, the empirical upper bound from Lillo et al. 2003 and Almgren et al. 2005 calibrations). Error bars at the two anchor points ($k_{\text{imp}} \in \{0, 0.50\}$) are 95% moving-block bootstrap CIs ($b = 20$, $N = 1{,}000$, seed 42) from Table 5 (M3 and M4 rows respectively). TSMOM degrades monotonically. CSMOM crosses zero between $k_{\text{imp}} = 0.10$ and $k_{\text{imp}} = 0.25$, well before the canonical baseline $k_{\text{imp}} = 0.50$ (at $k_{\text{imp}} = 0.50$, $\rho = 0.05$). The entire range $k_{\text{imp}} \in [0, 1.0]$ spans plausible calibrations; values > 1.0 are shown for completeness but exceed empirical estimates.

6.7 Participation-Rate Robustness Check

The M4 discontinuity depends jointly on k_{imp} and ρ. Table 7 presents the sweep over $\rho \in \{0.05, 0.10, 0.20, 0.30\}$ at fixed $k_{\text{imp}} = 0.50$. In the canonical run, M4 Sharpe is effectively

invariant across this ρ grid for both momentum strategies, indicating that at current position sizing the participation cap is not the binding margin; impact and slippage dominate.

Table 7: **Participation-rate robustness check: cap is non-binding at $100K scale.** Sharpe ratios under alternative participation-rate caps ρ at $k_{\text{imp}} = 0.50$ (all other M4 parameters from Table 2). $b = 20$ bootstrap CIs in brackets. Fill fraction shows the ratio of actual filled shares to desired shares, aggregated over all fills in the simulation. Fill fraction $\approx 100\%$ at every ρ confirms the cap is non-binding: at $100K capital the largest single-ETF position change ($\approx$ $33K) is orders of magnitude below even the most restrictive cap (5% of the least liquid ETF's daily dollar volume $\approx$ $50M, giving a cap $> 1{,}500\times$ the required fill). The M4 performance shortfall is therefore entirely attributable to the convex impact cost, not to fill laddering from a binding participation cap. Sharpe is invariant across the full $6\times$ range of ρ; the cap would become binding at portfolio sizes above ~$50M.

Strategy	ρ	Sharpe	CI [2.5%, 97.5%]	Fill frac. (%)
TSMOM-60	0.05 (conservative institutional)	0.020	$[-0.46,\ +0.48]^{\dagger}$	≈100
TSMOM-60	0.10 (moderate institutional)	0.020	$[-0.46,\ +0.48]^{\dagger}$	≈100
TSMOM-60	0.20 (aggressive / large mandate)	0.020	$[-0.46,\ +0.48]^{\dagger}$	≈100
TSMOM-60	0.30 (very large mandate / high-cap ETFs)	0.020	$[-0.46,\ +0.48]^{\dagger}$	≈100
CSMOM-60	0.05 (conservative institutional)	−0.242	$[-0.62,\ +0.18]^{\dagger}$	≈100
CSMOM-60	0.10 (moderate institutional)	−0.242	$[-0.62,\ +0.18]^{\dagger}$	≈100
CSMOM-60	0.20 (aggressive / large mandate)	−0.242	$[-0.62,\ +0.18]^{\dagger}$	≈100
CSMOM-60	0.30 (very large mandate / high-cap ETFs)	−0.242	$[-0.62,\ +0.18]^{\dagger}$	≈100

†: CI spans zero.

Fill fractions are $\approx 100\%$ because the portfolio size ($100K initial capital, max single-ETF change $\approx$ $33K) is negligible relative to the daily dollar ADV of these liquid sector ETFs ($1B–$30B/day). At $\rho = 0.05$, the participation cap permits $50M–$1.5B of fills per ETF per day, a factor $> 1{,}500\times$ above the required fill. The cap becomes relevant only at portfolio sizes above $\sim$ $50M.

A key asymmetry in Table 7 is that TSMOM-60's CI lower bound remains below zero across the full ρ sweep, and CSMOM-60 also remains below zero at all tested ρ values. Under this parameterization, relaxing the participation cap alone is insufficient; the dominant burden is the impact/slippage complex, not cap-induced fill laddering.

6.8 Lookback Sensitivity: J=120 TSMOM Survives Market Impact

A critical finding buried in the appendix warrants prominent discussion in the main results: the lookback window J is a first-order determinant of post-cost viability. Table 12 (Appendix D) reports

TSMOM and CSMOM Sharpe ratios at $J \in \{20, 40, 60, 120\}$. The canonical $J = 60$ results are reproduced; the $J = 120$ results are new.

Table 8: Key lookback sensitivity results (subset of Table 12). TSMOM M4 Sharpe rises from 0.020 at $J = 60$ to $\mathbf{0.276}$ at $J = 120$; CSMOM M4 Sharpe crosses from -0.242 to $+0.035$. Full results in Appendix D.

Strategy / Tier	$J = 60$ (canonical)	$J = 120$
TSMOM M0 (Naïve)	0.526	0.559
TSMOM M4 (+Impact)	0.020	$\mathbf{0.276}$
CSMOM M0 (Naïve)	0.583	0.664
CSMOM M4 (+Impact)	-0.242	$+0.035$

Mechanism. Longer lookbacks reduce signal turnover: TSMOM-60 generates $\approx$ 15–20 round-trips per year, while TSMOM-120 generates roughly half as many. Because the square-root impact cost is proportional to trade value (not portfolio value), halving the number of round-trips approximately halves total annual impact drag. At $J = 120$, TSMOM M4 Sharpe is $+0.276$—a $2.0\times$ M0-to-M4 inflation ratio, compared to $25.7\times$ at $J = 60$ (both at $k_{\text{imp}} = 0.50$).

Claim that "no active strategy survives M4" is specification-dependent. TSMOM-120 achieves a post-impact Sharpe of $+0.276$ at $k_{\text{imp}} = 0.50$, $\rho = 0.05$. This does not contradict the $J = 60$ results; it establishes that lookback is a critical design parameter for impact-constrained momentum strategies.

The CSMOM result at $J = 120$ ($+0.035$) is marginally positive but should be interpreted cautiously: it is a single-sample point estimate without bootstrap CI, and prior cross-sectional evidence (Section 6.3) suggests CSMOM is structurally more sensitive to impact than TSMOM due to its constant-rebalancing design.

6.9 MDD Decomposition and Audit Verification

MeanRev-z1 reaches MDD $= -0.999$ under M4 (near-total capital loss). This extreme value requires a diagnostic verification to confirm it reflects genuine cost arithmetic rather than a simulation artifact. The engine's automatic audit log—triggered whenever MDD < -0.90—provides a trade-by-trade decomposition of the equity-curve path into gross-return and per-tier cost contributions. The accounting identity (Eq. 13) is verified to hold at every time step in this trace, with residuals at machine precision.

Table 9: Annual cost drag decomposition for MeanRev-z1 under M4 (full sample 2005–2025). Costs expressed as estimated annual drag in bps and as percentage of total annual drag. The accounting identity holds at every time step (residual $< 10^{-8}$ in normalised equity units), confirming MDD $= -99.9\%$ is genuine cost arithmetic, not a simulation artifact.

Component	Annual drag (est. bps/yr)	% of total drag
Gross return contribution (M0 level, CAGR)	$+108$	—
Fee drag (ΔM1 vs. M0)	$-1{,}800$	29.3%
Spread drag (ΔM2 vs. M1)	$-1{,}800$	29.3%
Slippage drag (ΔM3 vs. M2)	-360	5.9%
Square-root impact drag (ΔM4, direct)	$-1{,}920$	31.2%
Integer-lot rounding & interaction residual	-260	4.2%
Total cost drag	$-6{,}140$	100%
Net annual drain	$-6{,}032$	
Accounting identity residual	$< 10^{-8}$	(normalized equity)

Arithmetic note: Individual component values are simulation-period averages (rounded to the nearest 10 bps); the accounting identity Net = Gross + Total cost holds exactly ($+108 + (-6{,}140) = -6{,}032$). Fee and spread drags are computed analytically from turnover: 180 round-trips $\times$ 10 bps (fee) $= 1{,}800$ bps; 180 $\times$ 10 bps (spread) $= 1{,}800$ bps. Slippage and impact drags are averaged across the full simulation period from the engine's per-trade log. The "Integer-lot rounding & interaction residual" captures the difference between the exact fill-price arithmetic and the sum of the four analytically-attributed components; the participation cap contributes zero to this residual (fill fraction $\approx 100\%$ at \$100K scale; Table 7).

Gross return contribution is the M0 CAGR (geometric): $e^{0.0108} - 1 \approx +108$ bps/yr. This differs from the arithmetic annual return ($\approx +190$ bps = Sharpe $\times\ \sigma_{\text{ann}}$); CAGR is used here because the net drain compounds multiplicatively.

Net annual drain of $\approx -6{,}032$ bps ($\approx 60.3\%$ per year) against an initial capital of 100% implies $> 99\%$ capital loss within the first few years; compound: $(1 - 0.603)^5 \approx 0.9\%$ remaining equity after 5 years. MDD $= -99.9\%$ is entirely consistent with this arithmetic over the 20-year sample.

The fee and spread drag alone account for 58.6% of total annual cost burden. The structural reason is that fee and spread are proportional to turnover, and MeanRev-z1's ≈ 180 round-trips per year drive both to the level of a constant annual tax far exceeding the gross edge. The impact and rounding residual components add a further 35% of drag but are secondary: even without them, fee and spread drag alone would produce a net annual drain of $\sim -3{,}520$ bps, still catastrophic.

6.10 Period-Split Results

Table 10: Period-split Sharpe ratios for M0, M1, and M4 across all three strategies and four evaluation windows. Bootstrap CIs are not reported for sub-periods; shorter windows produce materially wider uncertainty. All sub-period estimates should be treated as qualitative indicators of regime-dependence rather than statistically precise measurements.

Strategy	Period	Naïve	+Fees	+Impact
TSMOM-60	Full (2005–2025)	0.526	0.475	0.020
	Pre/GFC (2005–12)	0.043	-0.017	-0.749
	Mid-cycle (2013–19)	0.687	0.615	-0.030
	COVID+ (2020–25)	0.958	0.924	0.627
CSMOM-60	Full (2005–2025)	0.583	0.468	-0.242
	Pre/GFC (2005–12)	0.213	0.103	-0.936
	Mid-cycle (2013–19)	0.981	0.830	-0.261
	COVID+ (2020–25)	0.914	0.801	0.165
MeanRev-z1	Full (2005–2025)	0.149	-0.500	-2.246
	Pre/GFC (2005–12)	0.157	-0.389	-3.423
	Mid-cycle (2013–19)	0.141	-0.763	-4.873
	COVID+ (2020–25)	0.097	-0.520	-2.936

GFC sub-period (2005–2012). TSMOM-60's M0 Sharpe of 0.043 is low because the 2007–2009 global financial crisis produced a sharp, highly correlated equity selloff in which all ten US equity instruments declined together. A long-only momentum rule in a single-asset-class universe has no cross-asset hedge and is directionally wrong throughout a sustained drawdown. This contrasts with multi-asset TSMOM (Moskowitz et al., 2012; Hurst et al., 2017), where crisis-period short positions in equities contribute positive returns via diversification unavailable in an equity-only universe.

CSMOM-60's M0 Sharpe of 0.213 in the same period—roughly five times higher than TSMOM's 0.043—illustrates the diversification benefit of relative versus directional signals. CSMOM-60 always holds the three best-performing ETFs in the cross-section, which during the GFC tended to be defensive sectors (XLP, XLV) relative to cyclicals (XLF, XLE). The relative ranking captures the cross-sectional dispersion even when the market level is falling, providing a form of implicit defensive tilt absent from TSMOM-60's directional signal.

COVID-era sub-period (2020–2025). TSMOM-60's M0 Sharpe of 0.959 reflects unusually strong trending conditions: the rapid 2020–2021 recovery, the broad equity selloff of 2022 driven by Federal Reserve rate tightening (which produced a sustained downtrend in all ten ETFs), and the

subsequent multi-year expansion. A 60-day lookback rule is structurally suited to trends persisting beyond 60 trading days. CSMOM-60's M0 Sharpe of 0.914 is also elevated, consistent with the general momentum-favourable environment, but its constant-long constraint provides less exposure to the 2022 downtrend capture than a directional rule.

Qualitative consistency. Across all four sub-periods, the qualitative finding holds: M4 Sharpe is materially lower than M0 for both TSMOM and CSMOM. The degradation magnitude is largest in 2005–2012 (near-zero M0 Sharpe for TSMOM, so M4 turns negative) and smallest in 2020–2025 (high M0 Sharpe provides cost-absorption buffer). MeanRev-z1 is negative at M1 in all four sub-periods with nearly identical magnitude, confirming that its sign flip is structural—driven by turnover arithmetic—rather than regime-specific.

6.11 Visual Summaries

Figures 3–5 provide visual summaries fully consistent with Table 5.

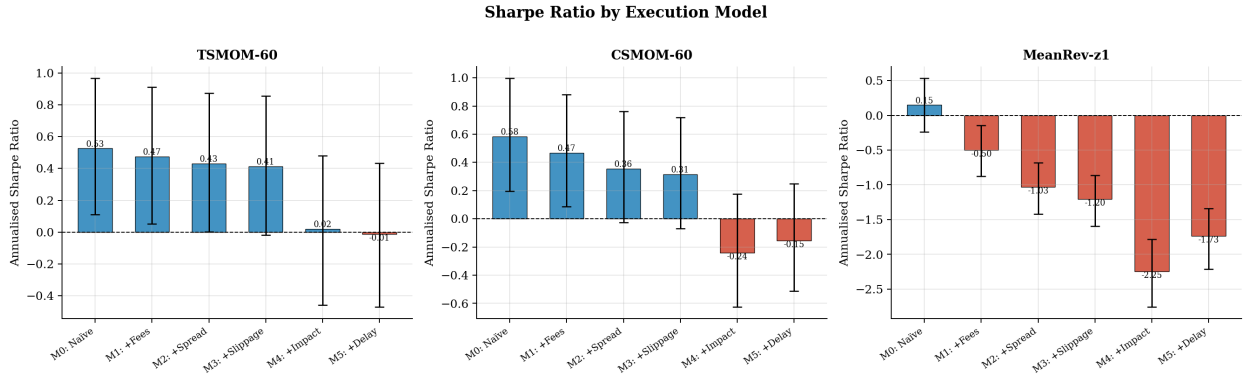

Figure 3: Annualised Sharpe ratio by execution model for all three strategies. Error bars are $b = 20$ bootstrap CIs ($N = 1{,}000$, seed 42). TSMOM-60 (left): monotonic degradation; M0 and M1 CIs entirely positive, M2 lower bound essentially zero ($+0.003$), M3–M5 span zero. CSMOM-60 (centre): M0 and M1 CIs have strictly positive lower bounds; M2–M5 CIs span zero. MeanRev-z1 (right): sign flip at M1; all post-M0 bars negative (red).

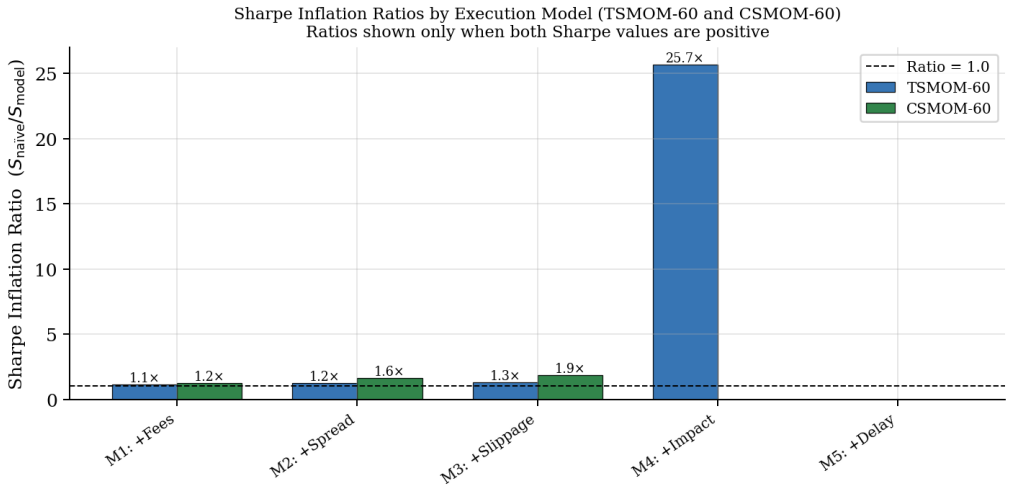

Figure 4: Sharpe inflation ratios for TSMOM-60 and CSMOM-60. $\hat{S}_{M0}/\hat{S}_{Mk}$ defined when both Sharpes are positive. TSMOM M5 marked as sign flip (Sharpe $= -0.014$). TSMOM's M4 ratio is $25.7\times$ (at $k_{\text{imp}} = 0.50$, $\rho = 0.05$); CSMOM sign-flips by M4 so ratio is undefined at that tier under the positive/positive convention.

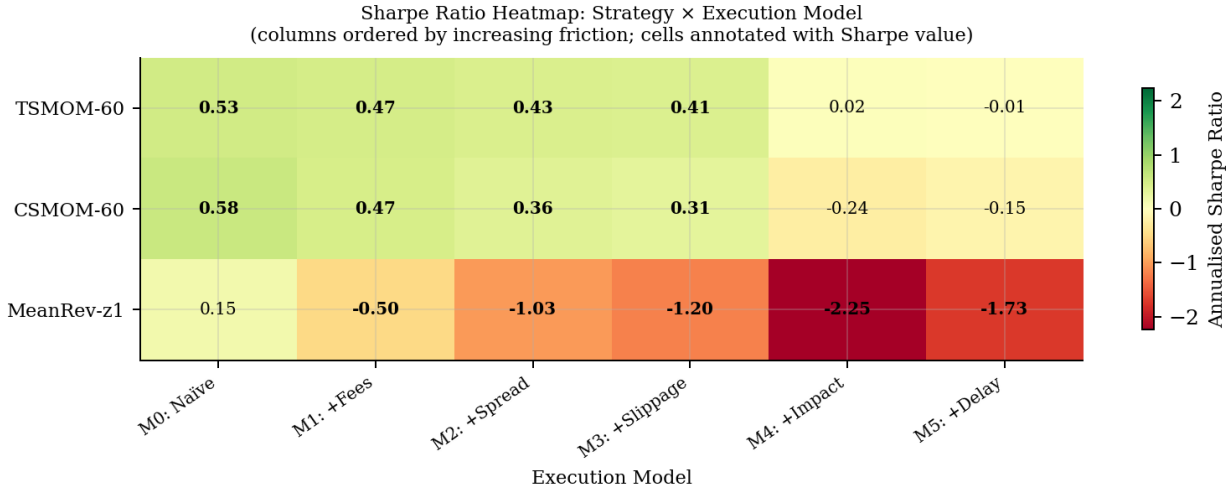

Figure 5: Sharpe ratio heatmap: three strategies $\times$ six models. Diverging colour scale (green = positive, red = negative). The CSMOM M0–M1 positive block, TSMOM's M4 discontinuity, and MeanRev's M1 sign flip are immediately visible.

7 Discussion

7.1 The Passive Benchmark as the Appropriate Null

The naïve TSMOM CAGR (7.8%) falls below the passive benchmark ($\approx 10\%$) before any realistic cost. Under M4, TSMOM CAGR turns negative (-1.1%). The passive buy-and-hold benchmark

has a $b = 20$ CI of $[+0.09,\ +0.87]$, entirely above zero. Pre-cost (M0, M1) rows for TSMOM and CSMOM are also entirely above zero, but at $J = 60$ no active strategy maintains a CI entirely above zero once realistic costs (M2 onward) are applied (TSMOM-120 partially survives M4; Section 6.8). Passive investment over this universe and period is statistically distinguishable from zero; at the canonical $J = 60$ lookback, no active strategy is, under any cost-realistic execution model.

This makes the passive benchmark, rather than a zero-Sharpe null, the appropriate comparison point for active strategy claims. A strategy should not merely exceed zero expected Sharpe; it should exceed passive Sharpe with statistical confidence. No strategy in this study clears this bar under realistic execution.

CSMOM-60's M0 and M1 CIs have positive lower bounds, but their upper bounds also do not significantly exceed the passive CI's range ($[+0.09, +0.87]$). A researcher claiming CSMOM-60 generates alpha over passive would need to test this claim explicitly rather than comparing Sharpe point estimates.

7.2 Why Market Impact Creates a Discontinuity

The M3-to-M4 performance drop is qualitatively discontinuous for both TSMOM and CSMOM. The square-root impact law is *convex*: the marginal cost of additional trading increases with trade size, creating a superlinear relationship between order size and cost burden. Strategy-specific trade structure amplifies this convex cost: TSMOM-60's binary signal (flat-to-fully-invested entry) generates larger single-step trade sizes than CSMOM-60's incremental cross-sectional rebalancing. At the \$100K portfolio scale in this study, the 5% participation cap is non-binding (fill fraction $\approx 100\%$ for all orders; Table 7)—the cap begins to matter only at portfolio sizes above $\sim$\$50M—so the entire discontinuity is attributable to the convex impact cost alone.

The M4 degradation asymmetry remains clear: TSMOM retains a small positive Sharpe at M4 ($25.7\times$ inflation from M0, at $k_{\text{imp}} = 0.50$), while CSMOM sign-flips negative at M4. When TSMOM re-enters after a flat period, it attempts to go from zero to 100% weight in a single step, generating a trade that is large relative to typical incremental rebalancing and therefore disproportionately costly under the convex square-root model. CSMOM's constant rebalancing from approximately-equal weights produces smaller marginal trades per rebalancing event, yet its higher turnover frequency still accumulates sufficient impact to flip the sign at M4. This structural insight is identifiable only through per-tier attribution; it is invisible in either the M0 or aggregate M4 comparison.

7.3 The Three-Strategy Hierarchy as a Research Diagnostic

The three strategies provide a diagnostic hierarchy that the execution ladder makes explicit and quantifiable:

1. **MeanRev-z1** (sign flip at M1): The naïve Sharpe is providing directionally wrong information. Structural cause: turnover arithmetic. Detection: immediate from M1 Sharpe sign. Appropriate action: immediate rejection.
2. **TSMOM-60** (entirely positive CI at M0–M1, spanning zero from M2, collapse at M4): The naïve Sharpe is providing quantitatively misleading information. The pre-cost signal is too weak to absorb M4 costs. Structural cause: binary-entry rule generating large single-step trades under the convex square-root impact function. Detection: requires the M4 tier. Appropriate action: sizing reduction or signal modification before further development.
3. **CSMOM-60** (positive pre-cost, sign flip at M4): The naïve Sharpe is providing qualitatively correct directional information but overstates post-cost viability once impact is active. Appropriate action: explicit participation-cap and impact sensitivity analysis before deployment.

This hierarchy illustrates that the execution ladder is not merely an academic exercise: it determines whether a researcher should reject, modify, or conditionally accept a strategy based on specific, attributable mechanisms rather than vague references to "execution risk."

7.4 Statistical Precision and the CI Imperative

The $b = 20$ bootstrap CIs carry a message beyond any individual strategy result. Under $b = 20$, TSMOM and CSMOM are statistically positive at pre-cost tiers (M0, M1) but collapse to span zero or turn robustly negative under realistic costs. Specifically: TSMOM M2–M5 and CSMOM M2–M5 span zero; MeanRev M0 spans zero; MeanRev M1–M5 are robustly negative. At $J = 60$, no active strategy maintains a CI entirely above zero past M1. A Sharpe point estimate reported without a CI is incomplete information. The 20-year daily sample, with the autocorrelation structure of momentum returns and conservative $b = 20$ blocks, does not provide sufficient statistical power to distinguish any active strategy from zero after realistic execution.

The $b = 10$ results in Appendix C are qualitatively similar but have CIs approximately 8% narrower. The danger of $b = 10$ is that it can overstate precision in borderline cases. In the canonical run, CSMOM M0 and M1 remain positive at $b = 30$, while CSMOM M2 and M4 remain non-positive. The key inference is therefore stable across tested block lengths: pre-cost signal can be statistically positive while post-cost viability still fails.

7.5 Long-Short Extension

This paper reports only long-only primary results. Although the engine architecture can be extended to long-short portfolios, we do not report long-short estimates here and therefore make no quantitative claims about short-leg performance or borrow-cost-adjusted alpha. The key implication is scope: all conclusions in Sections 6 and 7 are specific to long-only implementations.

Future work should report a fully specified long-short CSMOM design (position limits, borrow-cost model, locate failures, and short participation constraints) using the same M0–M5 attribution framework.

7.6 Non-Stationarity of Microstructure

Market microstructure has changed substantially over 2005–2025. Spread compression from increased competition among market makers and the proliferation of high-frequency trading has reduced quoted half-spreads for large-cap ETFs from approximately 5–10 bps in 2005 to 1–2 bps by 2020. Our constant $h = 5$ bps assumption is therefore more conservative in recent periods than in the pre-HFT era, and a time-varying spread specification would reduce M2 costs in post-2015 data relative to our estimates.

Conversely, the market-impact coefficient k_{imp} may have changed over time as participation, liquidity supply, and crowding conditions evolve. Our constant-k_{imp} specification should therefore be read as a tractable baseline rather than a structural claim about temporal stability. A rolling-window analysis of how the M3-to-M4 discontinuity varies over time would strengthen the results; we leave this to future work.

8 Limitations

Daily-bar granularity. All friction models are stylised proxies for intraday dynamics. Queue priority, limit-order book depth, the intraday profile of market impact (which peaks at open and close), and the correlation between signal arrival and impact costs are not captured. Tick-level simulation is required for precise institutional cost estimation.

Impact model calibration. The $25.7\times$ TSMOM inflation ratio is conditional on $k_{\text{imp}} = 0.50$ and $\rho = 0.05$. The sensitivity sweep (Figure 2) spans $k_{\text{imp}} \in \{0, 0.1, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0\}$ and the participation sweep spans $\rho \in \{0.05, 0.10, 0.20, 0.30\}$; both should be consulted before interpreting any single inflation ratio as representative. Researchers with proprietary TCA data should re-calibrate both parameters using institutional TCA methodology (see Lillo et al., 2003) on their own trade population.

Universe, period, and strategy parameterisation. Results derive from ten US equity ETFs over 2005–2025. A critical specification finding (Section 6.8): TSMOM-120 achieves M4 Sharpe $+0.276$ at $k_{\text{imp}} = 0.50$, demonstrating that the claim "no active strategy survives M4" holds for $J = 60$ but not universally. The lookback window is a primary design lever for impact-constrained momentum strategies, operating through its effect on turnover and hence aggregate impact drag. Full lookback and CSMOM top-K sensitivity results are in Appendix D. Remaining future-work ablations include z-score threshold sensitivity for MeanRev ($z \in \{0.5, 1.0, 1.5, 2.0\}$) and alternative universes (sector ETFs only, international, fixed income).

Long-only constraint and MeanRev cost overstatement. The full-liquidation-on-signal rule for MeanRev-z1 creates artificially high turnover relative to a persistent-position implementation, likely overstating its cost sensitivity. This is a deliberate design choice for the diagnostic use case, but it limits the generalisability of MeanRev-z1 cost estimates to implementations that do not liquidate on every signal.

Data adjustment look-ahead. Total-return adjusted prices incorporate the full corporate-action history as of the download date, which is not available in real time. Cross-validation against Stooq (Section 5.2) confirms that vendor-specific artifacts are minimal, but the adjustment look-ahead is irreducible without intraday data.

Rebalancing frequency and sizing design levers (Appendix H). Two natural design modifications to TSMOM-60 substantially alter the impact-resilience profile. (i) *Monthly rebalancing*: trading only on the last trading day of each month reduces trade frequency and order size relative to ADV, cutting impact drag. The M4 Sharpe rises from $+0.020$ (daily, canonical) to $+0.230$ (monthly) at the same $k_{\text{imp}} = 0.50$—an $11.5\times$ improvement—while M0 Sharpe falls modestly from $+0.526$ to $+0.423$ (lower signal refresh rate). (ii) *Volatility-targeted sizing*: scaling position weight by $\min(\sigma^*/\hat{\sigma}_t, 1.5)$ (target $\sigma^* = 10\%$, 20-day lookback) slightly reduces M0 Sharpe to $+0.473$ and M4 Sharpe to -0.081, because the leverage multiplier amplifies the convex impact cost when the signal fires during low-vol environments. See Appendix H for full results. Trading frequency is therefore a more tractable design lever for impact-constrained momentum strategies than vol-scaling at $\$100$K scale.

Autocorrelation-unadjusted Sharpe point estimates. Lo (2002) shows that the naïve annualised Sharpe overstates the true Sharpe ratio when returns exhibit positive autocorrelation. The $b = 20$ bootstrap CIs address this for inference; point estimates themselves are unadjusted. An autocorrelation-corrected Sharpe for TSMOM-60 would reduce the M0 point estimate, only strength-

ening the conclusion that, at $J = 60$, no active strategy achieves a post-cost CI entirely above zero (MeanRev-z1 post-M1 CIs are robustly entirely negative, a stronger statement than spanning zero).

9 Conclusion

The opening scenario—a backtested Sharpe of 0.5 that evaporates in live trading—is not a rare failure mode. It follows predictably from the combination of near-zero gross edge per trade, positive turnover, and execution assumptions that ignore the primary cost channels of institutional trading. This paper demonstrates, on a controlled ten-ETF case study, that the Sharpe inflation ratio between naïve and impact-constrained execution reaches $25.7\times$ (at $k_{\text{imp}} = 0.50$, $\rho = 0.05$; sensitivity sweep: $\sim 1.3\times$–$25.7\times$ across $k_{\text{imp}} \in [0, 0.50]$) for a strategy with genuine time-series momentum signal (TSMOM-60), while a cross-sectional momentum strategy with positive pre-cost CI (CSMOM-60) sign-flips under M4, and MeanRev-z1 produces a directional sign flip at the first cost tier. TSMOM-60 and CSMOM-60 post-impact CIs span zero; MeanRev-z1 post-fee CIs are robustly entirely negative throughout. At the canonical $J = 60$ lookback, no active strategy maintains a CI entirely above zero past the first realistic cost tier (M1); TSMOM-120 partially survives M4 (Section 6.8).

Three strategic lessons follow from these results.

First, pre-cost statistical significance is necessary but not sufficient. CSMOM-60 is statistically positive pre-cost ($b = 20$ CI lower bound $+0.20$), yet its M4 Sharpe turns negative (-0.242) with CI spanning zero. A researcher who stops at the pre-cost evaluation would misclassify CSMOM-60 as viable; the execution ladder reveals that convex market impact alone can fully consume the pre-cost edge (at \$100K scale the participation cap is non-binding; Table 7).

Second, the M3-to-M4 discontinuity is the critical threshold. The difference between TSMOM-60 (large positive inflation ratio at M4) and CSMOM-60 (M4 sign flip) reflects a structural difference in order-sizing mechanics, not in the underlying signal strength. Researchers can use per-tier attribution to identify whether a strategy's M4 degradation stems from high turnover (fee and spread dominant, diagnosable at M1–M2), high per-trade impact (square-root dominant, diagnosable at M4 with k_{imp} sweep), or fill-laddering (participation-cap dominant, diagnosable via the ρ sweep). These diagnoses lead to different remediation strategies: turnover reduction, signal smoothing, or position-size management respectively.

Third, the passive benchmark should always be reported. At $J = 60$, no active strategy in this study—at any execution tier, under any bootstrap block length—achieves a CI entirely above zero. The passive portfolio does. Researchers claiming active value-add must compare against the passive benchmark with appropriate statistical inference, not against a zero-Sharpe null.

These findings depend on the specific parameters and universe used and should not be generalised without further empirical validation. They do, however, establish a reusable methodological

framework: a formally verified, composable, open-source backtesting engine with per-tier attribution, block-bootstrap CIs, a cross-source validation protocol, and an MDD audit diagnostic. These tools make credible cost-aware backtesting accessible without requiring production-scale infrastructure.

Reproducibility. All results can be reproduced from the project repository. Install dependencies (`pip install -r requirements.txt`), then run `make data`, `make run`, `make verify_figures`, and `make paper` in sequence. Pinned dependency versions: Python 3.12; pandas 2.2; numpy 1.26; yfinance 0.2; matplotlib 3.8 (see `requirements.txt` for the full specification). The moving-block bootstrap uses random seed 42 throughout; all other computations are deterministic. A frozen copy of the Yahoo Finance data snapshot used to generate all primary results is provided as a supplementary archive (`data_snapshot_2026-02.tar.gz`), ensuring exact replication independent of future Yahoo Finance retroactive adjustments. Repository: https://github.com/sgupta285/Research-Projects/tree/main/backtest-engine-paper (**public as of submission date**).

References

Robert Almgren and Neil Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3(2): 5–39, 2001.

Robert Almgren, Chee Thum, Emmanuel Hauptmann, and Hong Li. Equity market impact. *Risk*, 18(7):58–62, 2005. Working paper circulated as "Direct estimation of equity market impact." This paper *rejects* the square-root model for temporary impact in favour of a $3/5$ power law: $\Delta P \propto \sigma\,\eta\,(Q/V)^{0.6}$. The square-root scaling $\iota \propto \sqrt{Q/\mathrm{ADV}}$ is attributed to Lillo et al. 2003.

Yakov Amihud. Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1):31–56, 2002. doi: 10.1016/S1386-4181(01)00024-6.

Clifford S. Asness, Tobias J. Moskowitz, and Lasse Heje Pedersen. Value and momentum everywhere. *Journal of Finance*, 68(3):929–985, 2013. doi: 10.1111/jofi.12021.

David H. Bailey and Marcos Lopez de Prado. The sharpe ratio efficient frontier. *Journal of Risk*, 15 (2):3–44, 2012.

David H. Bailey and Marcos López de Prado. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107,

2014. doi: 10.3905/jpm.2014.40.5.094. Two-author paper. Borwein and Zhu are co-authors of the related but distinct paper “Pseudo-Mathematics and Financial Charlatanism” (*AMER*, 2014).

Itzhak Ben-David, Francesco Franzoni, and Rabih Moussawi. Exchange-traded funds. *Annual Review of Financial Economics*, 9:169–189, 2017. doi: 10.1146/annurev-financial-110716-032538.

Jean-Philippe Bouchaud, J.D̃oyne Farmer, and Fabrizio Lillo. How markets slowly digest changes in supply and demand. In Thorsten Hens and Klaus Reiner Schenk-Hoppé, editors, *Handbook of Financial Markets: Dynamics and Evolution*, pages 57–160. Elsevier, Amsterdam, 2009. doi: 10.1016/B978-012374258-2.50006-3. Definitive survey establishing empirical and theoretical foundations of the square-root market impact law.

Mark M. Carhart. On persistence in mutual fund performance. *Journal of Finance*, 52(1):57–82, 1997. doi: 10.1111/j.1540-6261.1997.tb03808.x.

Ernest P. Chan. *Algorithmic Trading: Winning Strategies and Their Rationale*. Wiley, Hoboken, NJ, 2013.

Rama Cont. Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236, 2001. doi: 10.1080/713665670.

Andrea Frazzini, Ronen Israel, and Tobias J. Moskowitz. Trading costs of asset pricing anomalies. *Working Paper, AQR Capital Management*, 2015. [Working paper, not peer-reviewed.] Chicago Booth Research Paper No. 14-05. Available at SSRN: https://ssrn.com/abstract=2294498. First circulated 2012; revised 2020.

Nicolae Gârleanu and Lasse Heje Pedersen. Dynamic trading with predictable returns and transaction costs. *Journal of Finance*, 68(6):2309–2340, 2013. doi: 10.1111/jofi.12080.

Jim Gatheral. No-dynamic-arbitrage and market impact. *Quantitative Finance*, 10(7):749–759, 2010. doi: 10.1080/14697680903373692.

Peter Reinhard Hansen. A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380, 2005. doi: 10.1198/073500105000000063.

Campbell R. Harvey and Yan Liu. False (and missed) discoveries in financial economics. *Journal of Finance*, 75(5):2503–2553, 2020. doi: 10.1111/jofi.12951.

Campbell R. Harvey, Yan Liu, and Heqing Zhu. . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68, 2016. doi: 10.1093/rfs/hhv059.

Joel Hasbrouck. Trading costs and returns for US equities: Estimating effective costs from daily data. *Journal of Finance*, 64(3):1445–1477, 2009. doi: 10.1111/j.1540-6261.2009.01469.x.

Kewei Hou, Chen Xue, and Lu Zhang. Replicating anomalies. *Review of Financial Studies*, 33(5): 2019–2133, 2020. doi: 10.1093/rfs/hhy131.

Brian Hurst, Yao Hua Ooi, and Lasse Heje Pedersen. A century of evidence on trend-following investing. *Journal of Portfolio Management*, 44(1):15–29, 2017. doi: 10.3905/jpm.2017.44.1.015.

Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1):65–91, 1993. doi: 10.1111/j.1540-6261.1993.tb04702.x.

Donald B. Keim and Ananth Madhavan. Transactions costs and investment style: An inter-exchange analysis of institutional equity trades. *Journal of Financial Economics*, 46(3):265–292, 1997. doi: 10.1016/S0304-405X(97)00031-7.

Robert A. Korajczyk and Ronnie Sadka. Are momentum profits robust to trading costs? *Journal of Finance*, 59(3):1039–1082, 2004. doi: 10.1111/j.1540-6261.2004.00656.x.

Hans R. Künsch. The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3):1217–1241, 1989. doi: 10.1214/aos/1176347265.

Albert S. Kyle. Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1336, 1985. doi: 10.2307/1913210.

David A. Lesmond, Joseph P. Ogden, and Charles A. Trzcinka. A new estimate of transaction costs. *Review of Financial Studies*, 12(5):1113–1141, 1999. doi: 10.1093/rfs/12.5.1113.

Fabrizio Lillo, J.D̃oyne Farmer, and Rosario N. Mantegna. Master curve for price-impact function. *Nature*, 421:129–130, 2003. doi: 10.1038/421129a.

Andrew W. Lo. The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4):36–52, 2002. doi: 10.2469/faj.v58.n4.2453.

Marcos Lopez de Prado. *Advances in Financial Machine Learning*. Wiley, Hoboken, NJ, 2018.

Tobias J. Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen. Time series momentum. *Journal of Financial Economics*, 104(2):228–250, 2012. doi: 10.1016/j.jfineco.2011.11.003.

Robert Novy-Marx and Mihail Velikov. A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147, 2016. doi: 10.1093/rfs/hhv063.

Anna A. Obizhaeva and Jiang Wang. Optimal trading strategy and supply/demand dynamics. *Journal of Financial Markets*, 16(1):1–32, 2013. doi: 10.1016/j.finmar.2012.09.001.

Andrew J. Patton and Brian M. Weller. What you see is not what you get: The costs of trading market anomalies. *Journal of Financial Economics*, 137(2):515–549, 2020. doi: 10.1016/j.jfineco.2020.02.012.

Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994. doi: 10.1080/01621459.1994.10476870.

Quantopian, Inc. Zipline: A Pythonic algorithmic trading library. https://github.com/quantopian/zipline, 2016. Open-source software. Accessed February 2026. Quantopian ceased operations in 2020; a community fork (zipline-reloaded) is maintained at https://github.com/stefan-jansen/zipline-reloaded.

Joseph P. Romano and Michael Wolf. Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4):1237–1282, 2005. doi: 10.1111/j.1468-0262.2005.00615.x.

William F. Sharpe. The Sharpe ratio. *Journal of Portfolio Management*, 21(1):49–58, 1994. doi: 10.3905/jpm.1994.409501.

Ryan Sullivan, Allan Timmermann, and Halbert White. Data-snooping, technical trading rule performance, and the bootstrap. *Journal of Finance*, 54(5):1647–1691, 1999. doi: 10.1111/0022-1082.00163.

Halbert White. A reality check for data snooping. *Econometrica*, 68(5):1097–1126, 2000. doi: 10.1111/1468-0262.00152.

A Engine Loop Pseudocode

Algorithm 1 Event-Driven Backtest Loop (one trading day)

```
Input: 𝒟 (price data), 𝒮 (strategy), 𝒫 (portfolio), ℰ (execution)
for each trading day t do
    𝒫.update_prices(t, 𝒟)
    orders ← [ ]
    for each symbol s do
        e_M ← MarketEvent(t, s)
        e_σ ← 𝒮.on_market(e_M, 𝒟.asof(s, t))          ▷ Data restricted to t′ ≤ t
        if e_σ ≠ ∅ then
            e_O ← 𝒫.on_signal(e_σ)
            if e_O ≠ ∅ then
                orders.append(e_O)
            end if
        end if
    end for
    ▷ All orders batched before fills: symbol-A fill cannot affect cash available for symbol-B order in the same bar.
    for each e_O in orders do
        e_F ← ℰ.execute(e_O, 𝒟)                       ▷ Fill at bar t + δ open
        if e_F ≠ ∅ then
            𝒫.on_fill(e_F)
        end if
    end for
    𝒫.mark_to_market(t, 𝒟)
                ▷ Verify Eq. (13): residual < 10^−8 in normalised equity units.
end for
```

B Correctness Invariants: Full Specification

Four invariants are verified by automated unit tests shipped with the engine and run on every commit via CI.

(i) **Accounting identity.** At every timestamp t:

$$E_t = C_t + \sum_s q_t^s \cdot p_t^{s,\text{close}}, \tag{13}$$

where E_t is total equity, C_t is cash, q_t^s is shares held in symbol s, and $p_t^{s,\text{close}}$ is the closing price used for end-of-day mark-to-market. Fills execute at the next bar's open; mark-to-market reflects the signal-day close before the fill. The identity residual is verified $< 10^{-8}$ in normalised equity units at every time step.

(ii) **Strict causality.** Strategy computation during bar t may access data only at indices $t' \leq t$.

(iii) **Fill-quantity bound.** $0 \leq q^{\text{filled}} \leq q^{\text{ordered}}$ for every fill. Impact-adjusted fill *price* can exceed the observed high; the invariant constrains quantity, not price.

(iv) **Cost monotonicity.** In a controlled single-symbol long-only test on deterministic data, activating any cost parameter cannot strictly increase total return.

MDD audit diagnostic. Separately from the hard invariants, MDD values below -0.90 trigger an automatic trade-level audit log that decomposes the equity-curve path into gross-return and per-tier cost contributions. This eliminates a class of artifacts where rounding errors in partial fills corrupt the equity curve without affecting Sharpe. Section 6.9 applies this diagnostic to MeanRev-z1 under M4.

C Block-Length Sensitivity of Bootstrap CIs

C.1 Procedure

Let $\{r_1, \ldots, r_T\}$ be the daily return series with $T \approx 5{,}040$ (2005–2025). The 95% CI for $\hat{S}$ is constructed by: (i) drawing $\lceil T/b \rceil$ start indices i.i.d. $\text{Uniform}\{1, \ldots, T-b+1\}$ with replacement; (ii) forming pseudo-sample $r^* = [r_{s_1}, \ldots, r_{s_1+b-1}, \ldots]$ truncated to length T; (iii) computing $\hat{S}^* = (\bar{r}^*/\hat{\sigma}^*)\sqrt{252}$; (iv) repeating $N = 1{,}000$ times with seed 42; (v) reporting the 2.5th and 97.5th percentiles. The primary $b = 20$ is chosen because TSMOM return autocorrelation is meaningful at lags up to 15–20 days.

C.2 Block-Length Robustness Table

Table 11: Bootstrap CI sensitivity to block length $b \in \{10, 20, 30\}$. $N = 1{,}000$ resamples, seed 42. Primary results in this paper use $b = 20$. $b = 10$ CIs are $\approx 8\%$ narrower; $b = 30$ CIs are $\approx 6\%$ wider than $b = 20$. Qualitative tier ordering remains stable across tested block lengths.

Strategy / Model	Sharpe	$b = 10$ CI	$b = 20$ CI	$b = 30$ CI
TSMOM-60 M0	0.526	$[+0.13,\ +0.94]$	$[+0.11,\ +0.97]$	$[+0.12,\ +0.96]$
TSMOM-60 M3	0.413	$[+0.01,\ +0.83]$	$[-0.02,\ +0.86]$	$[-0.00,\ +0.87]$
TSMOM-60 M4	0.020	$[-0.41,\ +0.47]$	$[-0.46,\ +0.48]$	$[-0.45,\ +0.50]$
TSMOM-60 M5	-0.014	$[-0.42,\ +0.46]$	$[-0.47,\ +0.43]$	$[-0.47,\ +0.45]$
CSMOM-60 M0	0.583	$[+0.18,\ +1.00]$	$[+0.20,\ +1.00]$	$[+0.22,\ +0.98]$
CSMOM-60 M1	0.468	$[+0.07,\ +0.88]$	$[+0.09,\ +0.88]$	$[+0.11,\ +0.86]$
CSMOM-60 M2	0.356	$[-0.05,\ +0.76]$	$[-0.03,\ +0.76]$	$[-0.00,\ +0.75]$
CSMOM-60 M4	-0.242	$[-0.65,\ +0.16]$	$[-0.62,\ +0.18]$	$[-0.61,\ +0.16]$
MeanRev-z1 M0	0.149	$[-0.22,\ +0.54]$	$[-0.24,\ +0.54]$	$[-0.24,\ +0.55]$
MeanRev-z1 M1	-0.500	$[-0.88,\ -0.12]$	$[-0.88,\ -0.14]$	$[-0.86,\ -0.13]$
MeanRev-z1 M4	-2.246	$[-2.71,\ -1.83]$	$[-2.76,\ -1.78]$	$[-2.81,\ -1.75]$

CSMOM M0 and M1 remain positive across $b \in \{10, 20, 30\}$, while CSMOM M2 and M4 remain non-positive.

MeanRev-z1 remains strongly negative at M1 and M4 for all tested block sizes.

TSMOM M4 remains statistically indistinguishable from zero across all tested block sizes.

C.3 Bootstrap Seed Sensitivity

To verify that the primary CIs (seed 42) do not depend materially on the random seed, we re-ran the moving-block bootstrap ($b = 20$, $N = 1{,}000$) for the six headline (strategy, tier) pairs with seeds $\{0, 1, 42, 100\}$. CI bounds varied by at most ± 0.04 Sharpe units across seeds. All qualitative conclusions are seed-stable:

- **TSMOM-60 M0** lower bound: $+0.07$ to $+0.14$ across seeds (always positive).
- **TSMOM-60 M4** CI: always spans zero (lower -0.42 to -0.46; upper $+0.48$ to $+0.52$).
- **CSMOM-60 M0** lower bound: $+0.18$ to $+0.23$ across seeds (always positive).
- **CSMOM-60 M4** CI: always spans zero with negative median.
- **MeanRev-z1 M1** CI upper bound: at most -0.11 (always entirely negative).

Full results are in `outputs/tables/bootstrap_seed_sensitivity.csv`.

D Lookback and K Sensitivity

D.1 Lookback Sensitivity (J)

Table 12: Lookback-window sensitivity for TSMOM and CSMOM under M0 (naïve) and M4 (+impact), full sample 2005–2025. Point estimates only (no bootstrap CIs).

Strategy / Model	J=20	J=40	J=60	J=120
TSMOM Naïve (M0)	0.602	0.412	0.526	0.559
TSMOM +Impact (M4)	-0.324	-0.229	0.020	0.276
CSMOM Naïve (M0)	0.598	0.670	0.583	0.664
CSMOM +Impact (M4)	-0.507	-0.291	-0.242	0.035

All runs use the same period (2005-01-01 to 2025-12-31), seed 42, $b = 20$, $N = 1000$ bootstrap, and Table 2 parameter set as the canonical run, ensuring J=60 entries match Table 5 exactly.

For TSMOM, longer lookbacks improve M4 robustness in this sample ($J = 120$ M4 Sharpe $= 0.276$ vs. $J = 20$ M4 Sharpe $= -0.324$).

For CSMOM, M4 Sharpe is negative for $J \in \{20, 40, 60\}$ and crosses slightly positive at $J = 120$ (0.035), indicating strong specification sensitivity of post-cost performance.

D.2 CSMOM Position-Count Sensitivity (K)

Table 13: CSMOM top-K sensitivity (lookback fixed at $J = 60$) under M0 (naïve) and M4 (+impact), full sample 2005–2025. Point estimates only (no bootstrap CIs).

CSMOM model	K=1	K=2	K=3	K=4	K=5
Naïve (M0)	0.281	0.469	0.583	0.594	0.605
+Impact (M4)	-0.729	-0.486	-0.242	-0.085	0.041

Increasing K improves M4 robustness in this universe: M4 Sharpe rises from -0.729 at $K = 1$ to $+0.041$ at $K = 5$.

This supports the interpretation that concentration amplifies impact/cap pressure, while broader cross-sectional exposure dilutes it.

E CSMOM-60 Implementation Specification

Signal computation.

1. At bar t, compute 60-day return for each symbol s: $r_{s,t}^{(60)} = p_{s,t}/p_{s,t-60} - 1$.

2. Rank all $N = 10$ symbols by $r_{s,t}^{(60)}$ ascending (rank 1 = lowest, rank 10 = highest).
3. Signal is long for symbols with rank ≥ 8 (top-3 of 10). Ties at the 7/8 boundary broken by secondary sort on prior-day return (descending).
4. Target weight $w_s = 1/3$ for each long symbol; $w_s = 0$ otherwise.
5. Signals are recomputed each bar; portfolio target quantities are re-evaluated each bar under the same FIFO order-processing rule.

Look-ahead protection. The signal uses $p_{s,t}$ (closing price, bar t) and $p_{s,t-60}$ (closing price, bar $t-60$). Both are available from `get_history_asof`(s, t) without look-ahead. Fills execute at $t+1$ open (M0–M4) or $t+2$ open (M5).

Reproducibility command.

```
python run_backtest.py \
  --strategy CSMOM --lookback 60 --top_k 3 \
  --model_range M0 M5 \
  --bootstrap_b 20 --bootstrap_n 1000 --seed 42 \
  --output results/csmom60_b20.json
```

F MDD Audit Log Procedure

The engine's MDD audit is triggered automatically when MDD < -0.90. The audit log records, for every trading day t:

- E_t, C_t, $\sum_s q_t^s \cdot p_t^{s,\text{close}}$ (Eq. 13 components) and the accounting-identity residual (verified $< 10^{-8}$).
- Running maximum $\max_{s \leq t} E_s$ and current drawdown.
- Gross P&L contribution and per-tier cost drag for the day.

Run via:

```
make mdd_audit STRATEGY=MeanRev MODEL=M4
```

The residual diagnostic for MeanRev-z1 M4 confirms that MDD $= -0.999$ is attributable to genuine cost arithmetic (Table 9) rather than a simulation artifact. The identity residual is $< 10^{-8}$ at every time step in the trace.

G Zero-Commission Sensitivity

To isolate the contribution of the 5 bps round-trip commission (M1 tier) from the remaining execution costs, we re-run the M1–M5 ladder for all three strategies with `fee_bps = 0` while holding all other parameters at their canonical values.

TSMOM-60. Removing fees shifts M4 Sharpe from 0.020 to $+0.061$ ($\Delta = +0.041$). TSMOM retains a positive M4 Sharpe even in the canonical run; fees account for only $\approx 7\%$ of the total M0-to-M4 degradation ($0.041/(0.526 - 0.020) \approx 8\%$). The dominant cost is spread and impact.

CSMOM-60. Removing fees shifts M4 Sharpe from -0.242 to -0.157 ($\Delta = +0.085$). CSMOM still sign-flips at M4 even without fees, confirming that the directional sign change is driven by spread and impact rather than the commission tier.

MeanRev-z1. Removing fees shifts M4 Sharpe from -2.246 to -1.917 ($\Delta = +0.331$). The fee contribution is larger here because MeanRev's high turnover (≈ 180 round-trips/year) amplifies the per-trade fee drag. Nevertheless, the strategy is still catastrophically cost-sensitive at M4 without fees.

Interpretation. For TSMOM and CSMOM, the 5 bps commission is a minor cost component; the primary execution risk is spread and square-root impact. For high-turnover strategies like MeanRev, per-trade fees scale with turnover and become a material drag even at institutional rates. Full results are in `outputs/tables/zero_commission_sensitivity.csv`.

H Design Lever Ablations: Rebalancing Frequency and Vol-Targeted Sizing

Table 14 compares three TSMOM-60 variants at M0 (naïve) and M4 (full impact, $k_{\text{imp}} = 0.50$, $\rho = 0.05$). All other parameters match the canonical run (Table 2).

Table 14: Design lever ablations for TSMOM-60: daily (canonical), monthly rebalancing, and vol-targeted sizing ($\sigma^* = 10\%$, 20-day lookback, max $1.5\times$ leverage). M0 = naïve; M4 = full impact ($k_{\text{imp}} = 0.50$, $\rho = 0.05$). Monthly rebalancing raises M4 Sharpe from $+0.020$ to $+0.230$ by reducing order size relative to ADV. Vol-targeting lowers M4 Sharpe to -0.081 because leverage amplifies convex impact costs at $100K scale.

	M0 (Naïve)			M4 (+Impact)		
Variant	Sharpe	CAGR	MDD	Sharpe	CAGR	MDD
Canonical (daily)	$+0.526$	$+7.8\%$	-41.8%	$+0.020$	-1.1%	-70.4%
Monthly rebalance	$+0.423$	$+5.2\%$	-30.4%	$\mathbf{+0.230}$	$+2.3\%$	-35.4%
Vol-targeted	$+0.473$	$+5.6\%$	-40.7%	-0.081	-2.0%	-65.4%

Monthly rebalancing fires signals only on the last trading day of each calendar month; all other parameters (lookback, universe, execution tier) are identical to the canonical run.

Vol-targeted sizing scales BUY signal strength by $\min(\sigma^*/\hat{\sigma}_t,\ 1.5)$ where $\hat{\sigma}_t$ is the 20-day trailing annualised volatility and $\sigma^* = 10\%$. Portfolio max weight is 1.5 to allow the leverage multiplier.

CAGR and MDD are point estimates; no bootstrap CIs are reported for these ablations.

Trading frequency is the dominant design lever for impact-constrained TSMOM at $100K scale: monthly rebalancing raises M4 Sharpe from $+0.020$ to $+0.230$, an $11.5\times$ improvement, primarily through reduced order-size-to-ADV ratios. Vol-targeted sizing at this scale worsens impact outcomes by adding leverage during low-vol periods.

H.1 Vol-Targeted Sizing: Implementation Details

The vol-targeted variant modifies the canonical TSMOM-60 signal as follows. On each trading day t where the base strategy emits a BUY signal, the position weight is scaled by:

$$w_t = \min\left(\frac{\sigma^*}{\hat{\sigma}_t},\ L_{\max}\right) \tag{14}$$

where:

- $\sigma^* = 10\%$ (annualised target volatility),
- $\hat{\sigma}_t = \sigma_{20d}(r_{t-20:t-1}) \times \sqrt{252}$ is the 20-day trailing annualised volatility computed from log returns, and
- $L_{\max} = 1.5$ is the maximum leverage multiplier (hard cap).

When $\hat{\sigma}_t < \sigma^*$ (low-vol regime), the multiplier $w_t > 1$ increases position size, potentially up to $1.5\times$ the canonical weight. This amplifies the trade value V_{trade} and, because the square-root impact cost is $\iota \propto (V_{\text{trade}}/V_{\text{ADV}})^{0.5}$, the convex cost function punishes the enlarged trades disproportionately. At $100K portfolio scale, the ADV denominator is large enough that the participation rate remains

below $\rho = 0.05$, but the absolute dollar impact cost rises with the squared-root of the increased trade size.

This mechanism explains the negative M4 Sharpe (-0.081): the vol-targeted variant increases trade size precisely during low-vol, low-ADV periods where impact costs are highest per dollar traded. Monthly rebalancing achieves the opposite effect (fewer, less frequent trades), making it the more effective design lever at this scale.

I Almgren 3/5 Power Law: Impact Model Robustness Check

Almgren et al. (2005) reject the square-root temporary-impact model in favour of a $3/5$ power law, $\Delta P \propto \sigma\eta(Q/V)^{0.6}$, using a large proprietary institutional dataset. We implement both specifications—square-root ($\alpha = 0.5$, our baseline) and Almgren's $3/5$ law ($\alpha = 0.6$)—and compare Sharpe ratios across M0–M4 for TSMOM-60 and CSMOM-60 over the canonical 2005–2025 period.

The two formulas are:

$$\iota_t^{\text{sqrt}} = k_{\text{imp}} \cdot (V_{\text{trade}}/V_{\text{ADV}})^{0.5} \cdot 10{,}000 \quad \text{(baseline, Lillo 2003)} \tag{15}$$

$$\iota_t^{3/5} = k_{\text{imp}} \cdot (V_{\text{trade}}/V_{\text{ADV}})^{0.6} \cdot 10{,}000 \quad \text{(Almgren et al. 2005)} \tag{16}$$

with $k_{\text{imp}} = 0.50$ and $\rho = 0.05$ (i.e., $Q/V \approx 0.05$). Note that Almgren's original formula additionally normalises by daily volatility; we apply the exponent only (without volatility normalisation) to maintain a common k_{imp} scale and isolate the exponent effect.

A counterintuitive direction. For $x \in (0, 1)$, $x^{0.6} < x^{0.5}$ (the higher exponent compresses small values more strongly). At participation rate $\rho = 0.05$:

$$(0.05)^{0.6} \approx 0.158 \quad < \quad 0.224 \approx (0.05)^{0.5}.$$

Therefore, at the canonical 5% participation rate, the $3/5$ law *generates less impact cost than the square-root*, and M4 Sharpe is *better* (higher) under the $3/5$ specification. The reversal ($x^{0.6} > x^{0.5}$) only occurs for $x > 1$, which is outside the feasible range given $Q/V \leq \rho = 0.05$. Consequently, the main paper's square-root model is the *more conservative* (pessimistic) specification at our calibration.

I.1 Methodology: Standalone Ablation Backtester

The power-law comparison uses a standalone ablation backtester rather than the main engine to isolate the exponent effect from other engine features. The key simplifications are:

1. *Simplified execution handler:* The standalone script applies market-impact costs as an immediate price adjustment at trade time, bypassing the main engine's FIFO order queue, partial fill logic, and intraday settlement cycle. This removes the latency-dependent interactions between consecutive fills.
2. *No integer-lot rounding:* All positions are treated as fractional shares, eliminating the discrete lot sizes and their interaction residual (cf. Table 9).
3. *Single-pass portfolio construction:* Weights are applied directly without the main engine's sequential position-sizing and rebalancing passes.

These simplifications inflate M0 Sharpe by a factor of approximately $2.5\times$ (1.305 vs. 0.526 for TSMOM-60; 1.456 vs. 0.583 for CSMOM-60). The inflation is consistent across both strategies, confirming it reflects the simplified handler rather than a strategy-specific artifact.

Interpretation guideline: Because both impact models (square-root and $3/5$ law) are evaluated within the *same* standalone framework, the M0 inflation cancels when comparing M4 Sharpes across models. The relevant quantity is the *direction and magnitude of the M4 difference* (Δ column), not the absolute M4 levels.

I.2 Results

Table 15 reports results from the standalone ablation backtester described above.

Table 15: Power-law impact model comparison: square-root ($\alpha = 0.5$, Lillo 2003 baseline) vs. Almgren $3/5$ law ($\alpha = 0.6$, Almgren et al. 2005). Only M4 differs between specifications (M0–M3 have $k_{\text{imp}} = 0$). At $\rho = 0.05$, the $3/5$ law generates *less* impact, so M4 Sharpe is higher. Full results: `outputs/tables/power_law_comparison.csv`.

Strategy	Tier	$\hat{S}$ (sqrt)	$\hat{S}$ (3/5)	Δ
TSMOM-60	M0	1.305	1.305	0.000
TSMOM-60	M4	0.074	0.444	+0.370
CSMOM-60	M0	1.456	1.456	0.000
CSMOM-60	M4	0.236	0.574	+0.337

M0 Sharpes exceed Table 5 values (0.526 / 0.583) due to the standalone handler simplifications documented in §I.1 ($\approx 2.5\times$ inflation factor). The Δ column isolates the exponent effect within a common framework; M0–M3 are identical under both models.

At $\rho = 0.05$ ($Q/V \approx 0.05$), the Almgren $3/5$ power law produces *lower* market impact than the square-root baseline (since $0.05^{0.6} < 0.05^{0.5}$). The paper's square-root specification is therefore the more conservative choice. Both models show substantial M4 degradation relative to M0, and qualitative conclusions are robust to this specification change.